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# Loan guarantees and credit supply☆

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ABSTRACT

The efficiency of federal lending guarantees depends on whether guarantees increase lending supply or simply act as a subsidy to lenders. We use notches in the guarantee rate schedule for Small Business Administration (SBA) loans to estimate the elasticity of bank lending volume to loan guarantees. We show significant bunching in the loan distribution on the side of the size threshold that carries a more generous loan guarantee. The excess mass implies that increasing guarantee generosity by one percentage point of loan principal would increase per-loan lending volume by $19,000. Excess mass increases in periods with guarantee generosity, and placebo results indicate that the effect disappears when the guarantee notch is eliminated.

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## 1. Introduction

Indirect government loan guarantees reimburse unrecovered dollars to private lenders and are an increasingly common type of credit subsidy. In 2019 alone, $1.4 out of the $1.5 trillion dollars in projected federal credit assistance came in the form of loan guarantees, with a projected subsidy value of $37.9 billion.[1] This paper studies how private lenders respond to federal loan guarantees. In markets affected by asymmetric information and credit rationing, government loan guarantees can increase aggregate welfare if they restore lending to an efficient level (see, e.g., Gale, 1991; Stiglitz and Weiss, 1981; Smith, 1983; Mankiw, 1986). Whether this occurs is ultimately an empirical question and depends in part on the responsiveness of lenders to the guarantee. Whether federal guarantee programs have any effects on increasing access to credit, or simply act as a subsidy to lenders, depends on the elasticity of credit provision to the loan guarantee. If credit supply is inelastic, guarantees will not increase the level of borrowing and will simply reimburse

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[1] See the report "Fair-value estimates of the cost of federal credit programs in 2019" by the Congressional Budget Office Research (https://www.cbo.gov/publication/55278).

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lenders on their losses. In this case, government loan guarantees can also crowd out more efficient private borrowing and encourage excessive risk-taking. Despite the large and growing volume of federally guaranteed debt, there remains relatively little work exploring the effects of federal guarantees on lending.

In this paper, we focus on how guarantees affect the supply of credit to small businesses. Credit constraints are well-known barriers to growth for small firms, and these problems are especially severe given imperfect information and a lack of collateral (see, e.g., Fazzari et al., 1988; Petersen and Rajan, 1994; Petersen and Rajan, 1995; Kaplan and Zingales, 1997; Barrot, 2016). Prior work has shown that these programs can alleviate barriers to entrepreneurship (Lelarge et al., 2010). We employ data from the Small Business Administration (SBA), the government agency tasked with providing assistance to small businesses. Specifically, we utilize data on loans originated under the 7(a) loan program. Under the SBA 7(a) loan program, a portion of loans from commercial lenders are insured against losses from defaults. Loans of up to $150,000 carry a higher maximum guarantee rate than loans larger than $150,000. This feature of the federal guarantee program leads to sharply different levels of risks for lenders originating loans above and below the threshold.

We employ a bunching estimator to measure the excess mass at the threshold and use this to estimate the elasticity of loan supply to the guarantee rate. We use a simple model to translate the observed excess borrowing at the mass into an elasticity of credit supply. The degree of bunching identifies the elasticity of lending supply to the guarantee; if lending supply is inelastic and lenders do not adjust loan size in response to the guarantee, we will not observe bunching. On the other hand, if lending supply is highly elastic, we will observe bunching, as a significant number of loans will be moved to the side of the threshold with higher guarantees.

We find significant bunching directly below the threshold, which translates to a highly elastic lending supply response to loan guarantees. Interpreted in dollar magnitudes, this means that a one percentage point change in the guarantee net subsidy rate (expressed as a percentage of loan principal) generates $19,054 in additional lending. Guarantee thresholds change over time, and we find that the observed bunching is stronger in years when guarantee amounts across the threshold are higher. We find that the elasticity varies slightly from year to year, and consistent with optimization frictions, we find smaller elasticities in years immediately after guarantee notches have changed. Moreover, the guarantee notch was eliminated during a two-year period from 2009 to 2010, as part of the American Recovery and Reconstruction Act (ARRA). During this period, we find no excess mass across the threshold, which serves as a placebo test to rule out the possibility that alternative factors might be changing across the threshold and driving our results.

The validity of the bunching estimate relies on two key assumptions: first, the counterfactual distribution is smooth in the absence of a notch, and second, there exists a well defined marginal buncher. Consistent with our identifying assumptions, we find no excess mass in years when the guarantee notch is eliminated, making it unlikely that other factors are changing at the threshold. Additionally, we find no differences in loan terms around the threshold: interest rates, maturities, revolving loan percentages, and charge-off percentages appear similar at or near the notch. We rule out several alternative explanations and threats to identification. According to SBA rules, lenders are only able to issue one loan to borrowers that have exhausted other borrowing options. We confirm in the data that lenders are not issuing multiple loans to the same borrower to take advantage of guarantees. We also find no difference in interest rates at or around the threshold, which is likely due to a particular institutional detail–the majority of loans in this program have binding interest rate caps, and thus there is very little room to vary the interest rate. This supports a channel from distributional responses driven by supply, rather than demand, side forces.

Our analysis sheds light on an ongoing policy debate regarding the efficiency of government loan guarantees. Proponents of lending guarantee programs argue that guarantees provide credit to borrowers that would otherwise be unable to access funds. However, opponents of loan guarantee programs contend that these programs simply serve as a subsidy to lenders. Major pushes to shut down the SBA were undertaken by the executive branch and Congress in 1984 and 1996, with pressure continuing into the 2010s. For example, a 2012 Wall Street Journal op-ed noted that “Congress created the Small Business Administration in 1953 to fix a specific problem: Lenders allegedly were turning away large numbers of small businesses that, if given a loan, would generate untapped economic growth. It is questionable whether this problem ever existed... The SBA loan program is best understood as a subsidy to banks. Borrowers apply to an SBA-certified bank. The SBA guarantees 75% to 85% of the value of loans made in the flagship program. The banks then boost their earnings by selling the risk-free portion of the loans on a secondary market.” See the Congressional Budget Office (CBO) for a discussion of proposals to eliminate the SBA. As well as being important to policy, the effect of loan guarantees on the supply of funds is a key parameter in many models of the effects of guarantees. For example, Smith (1983) notes that “To be effective, it must be demonstrated that there is some impact of these policies on supply elasticities of credit.” Gale (1991) states that cäPerhaps the single most important and controversial parameter is the elasticity of supply of funds.” Finally, Lucas (2016) notes that “The elasticity of credit supply affects the extent to which additional borrowing in government credit programs is offset by reductions in private borrowing.”

We inform this debate by focusing on the guarantee program that serves as a major source of small business financing in the United States. The SBA is an important source of small business financing, with $25.4 billion in SBA-guaranteed loans made in 2018, mainly through the 7(a) program. This funding is typically provided to young firms at a critical point in the firm’s life cycle when they are unable to access other sources of capital. A number of well-known major companies secured SBA loans in early stages. These include Apple, FedEx, Nike, Intel, Under Armour, Whole Foods, and Chipotle.

This paper contributes to a body of work on federal lending subsidies and guarantees by estimating a key parameter from classic theory models. Despite the growing volume of federal lending in recent years, the area remains underexplored relative to other credit markets. Notable exceptions include Gale (1990), Gale (1991), Smith (1983), and Lucas (2016). To our knowledge, this is the first empirical paper to estimate how lending supply responds to federal loan guarantees. This literature largely focuses on calibrated models, and different papers use a wide range of estimates of the elasticity of credit supply to guarantee rates for calibrations.

Other work has focused on different aspects of government credit guarantees. La Porta et al. (2002) examine the effect of government ownership of banks, and find a positive correlation between government intervention and slower subsequent financial development that is consistent with government crowding out efficient private borrowing. Bertrand et al. (2007) examine the effect of the French Banking Act of 1985, which eliminated government subsidies to banks intended to help small- and medium-sized firms. Lelarge et al. (2010) study the effects of a French guarantee program on entrepreneurship. Atkeson et al. (2018) emphasize the role of government guarantees in bank valuation by arguing that the decline in banks' market-to-book ratio since the 2008 crisis is due to changes in the value of government guarantees. Kelly et al. (2016) show that government guarantees lower financial sector index prices.

Prior theory work has shown that under information asymmetries, government interventions in credit markets such as loan guarantees and loan subsidies can increase welfare (see, e.g., Stiglitz and Weiss, 1981; Mankiw, 1986; Greenwald and Stiglitz, 1986). More recent work by Hanson et al. (2018) has focused on tradeoffs between private and social costs, and Fieldhouse (2018) shows that housing policies subsidizing an expansion in residential mortgage lending crowd out commercial mortgages and loans. While in theory loan guarantees can increase welfare, whether this is true in practice is ultimately an empirical question. We show that private lending is indeed responsive to federal loans guarantees, suggesting that these programs have real effects beyond simply subsidizing lenders.

This paper also links to a literature on credit access for entrepreneurs and small firms. Financing constraints are well known to be a significant barrier to growth for small firms (see, e.g., Evans and Jovanovic, 1989; Whited and Wu, 2006; Rauh, 2006; Kerr and Nanda, 2010; Barrot, 2016; Adelino et al., 2017). A large body of work studies small enterprises' financial frictions and various policy responses. Petersen and Rajan (1994), Petersen and Rajan (1995), and Darmouni (2017) show that, for small firms, close ties with institutional lenders increases the availability of credit. Darmouni and Sutherland (2018) show that lenders to small firms are highly responsive to competitors' offers.

More recent work has focused on how federal programs can affect the supply of credit and entrepreneurship. Brown and Earle (2017) and Granja et al. (2018) study the SBA program and, respectively, find that access to credit has large effects on employment and the average physical distance of borrowers from banks' branch matters for ex-post loan performance. Barrot et al. (2019), Mullins and Toro (2017), and Gonzalez-Uribe and Wang (2019) study similar programs to stimulate small business lending in France, Chile, and the UK. Howell (2017) demonstrates that federal grants have large effects on future fundraising, patenting and revenue. This paper shows that the volume of small business lending is highly responsive to loan guarantees and loan guarantees can be a relatively low cost way to increase lending to small enterprises.

Beyond the use of these estimates directly for the growing literature on loan guarantees, our estimates and their implications for the supply of credit to small businesses are relevant for structural models of entrepreneurship and firm dynamics. For example, Evans and Jovanovic (1989) assume that the lending rate equals the borrowing rate, which implies that the supply curve of capital is not upward sloping over a wide range. Herranz et al. (2015) additionally assume that debt is provided by a risk-neutral competitive lender with an elastic supply of funds. Our estimates are also of use in terms of estimating the marginal value of public funds (e.g Hendren, 2014; Hendren, 2016), specifically in terms of the SBA program for welfare analysis. The marginal value of public funds maps the causal estimates of a policy change into welfare analysis by comparing the ratio of the beneficiaries' willingness to pay for a program with the net cost to government, in other words, cost-benefit analysis.

The remainder of this paper is organized as follows. Section 2 discusses institutional details on SBA loans and federal guarantees and describes the SBA data used in our analysis. Section 3 presents an illustrative model and discusses how our identification strategy is linked to this model. Section 4 introduces the bunching estimator and discusses the empirical approach. Section 5 presents the main results and demonstrates significant lending response to government guarantees. Section 6 discusses alternative explanations and presents placebo results. Section 7 concludes the paper and discusses avenues for further research.

## 2. Institutional background and data

Federal loan guarantee programs operate in a fashion similar to insurance contracts. Lenders pay a fee to the government, and in return, the government reimburses a portion of dollars that are charged off when a loan goes into default. Loan guarantees exist or have existed in several loan markets, such as in student, mortgage, and small business lending markets. In this paper, we focus on loan guarantees in the small business lending market. This section discusses the institutional details surrounding the SBA 7(a) program studied in our empirical analysis.

### 2.1. SBA 7(a) loans

The SBA is an independent federal government agency created in 1953 with the mission of providing assistance to small businesses. We focus on the lending program, designed to improve access to capital for young small businesses that may not be eligible to obtain credit through

traditional lending channels. The SBA lending programs are guarantee programs where the SBA guarantees a portion of loans originated by commercial lending institutions against losses from defaults rather than lending directly to qualifying borrowers. We focus on the SBA's flagship loan guarantee program, the 7(a) loan program.

SBA 7(a) guarantees consist of two components, a reimbursement rate and a fee. The reimbursement rate is the fraction of each dollar charged off that the bank receives back from the SBA, and the fee is the amount that the bank must pay to participate in the 7(a) program. There are several features of guarantee components that are relevant to this study. Most importantly, the maximum guarantee rate is based on a nonlinear size cutoff rule: loans up to $150,000 carry a maximum guarantee rate of 85%, which drops sharply to 75% for loans larger than $150,000. The guarantee fees also increase at the same threshold, making the overall guarantee less generous for loans larger than $150,000. We exploit this guarantee notch around $150,000 to identify our parameters of interest. Features of the SBA 7(a) program have remained relatively stable over the last decade, except during 2009–2010, when the SBA temporarily raised the guarantee rate on either side of the $150,000 threshold to 90% and waived fees with the signing of the ARRA of 2009. This time period provides a helpful placebo test for our analysis since no lending response should occur in a year when there is no discrete change in the guarantee rate.

To qualify for a 7(a) loan, a borrower must meet several requirements. First, it must be a for-profit business that meets SBA size standards. Size standards vary by industry and are based on the number of employees or the amount of annual receipts ("total income" plus "the costs of goods sold"). In addition to the size requirement, a business must be independently owned and operated and not be nationally dominant in its field. The business must also be physically located and operate in the US or its territories. Last, small businesses must demonstrate the need for a loan by providing loan application history, business financial statements, and evidence of personal equity investment in the loan proposal.

To qualify, borrowers must exhaust other funding sources, including personal sources, before seeking financial assistance and be willing to pledge collateral for the loan.[2] SBA 7(a) loans are intended as a last resort, and to ascertain that borrowers cannot access credit elsewhere, lenders are required to conduct "credit elsewhere" tests. The SBA provides further information regarding credit elsewhere tests. In addition, Appendix Table A.3 shows the fraction of firms accessing multiple sources of credit in the 2003 Federal Reserve Survey of Small Business Finances (SSBF) that have loans from a government agency, including the SBA. The table indicates that very few firms that have SBA loans are accessing credit from multiple sources. Lenders are required to demonstrate that borrowers cannot obtain the loan on reasonable terms without the SBA guarantee and that the funds are not available from the resources of the applicant. The personal resources of any applicant who owns more than 20% of the small business are reviewed. The SBA monitors lenders' compliance with the credit elsewhere test through targeted reviews. Failure to comply with credit elsewhere tests can lead to the denial of a guarantee, exclusion from the lending program and other enforcement actions from the Office of Credit Risk Management.

The 7(a) loans are disbursed through private lending institutions. This loan submission and disbursement procedure depends largely on the lender's level of authority (i.e., delegated or non-delegated) provided by the SBA. The SBA conducts its own analysis of the application and approves the originating lender's decision to lend, which can be expedited depending on a lender's experience. In practice, SBA lenders have meaningful bargaining power over credit supply. In a typical case, a borrower requests a loan to a lender, and the lender decides whether the SBA loan would be suitable for a given borrower upon reviewing the borrower's background. Given that lenders cannot provide more than one loan to a single borrower such that the SBA-guaranteed loan is secured with a junior lien position, lenders have incentives to retain this bargaining power and to be selective in choosing borrowers.

Note that the reimbursement rate and fees are typically determined by an Office of Management and Budget (OMB) model, vary from year to year, and have been changed through legislation such as the ARRA. The CBO notes that "One of the SBA's goals is to achieve a zero subsidy rate for its loan guaranty programs," which entails generating revenue from fees and recoveries to offset program costs. In practice, the SBA is sometimes successful and sometimes not in terms of achieving a zero subsidy rate. Between 2007 and 2009, and between 2014 to the present day, the program operated at zero subsidy. The CBO report on the SBA 7(a) loan guaranty program provides further detail regarding the goals and subsidy rates of the program.

### 2.2. Data

We obtain the 7(a) loan data from the SBA. The SBA requires all participating lenders in the 7(a) program to submit loan applications (Forms 1919 and 1920) to the 7(a) Loan guaranty Processing Center ("LGPC") when they request a new loan. Delegated lenders must complete the form, sign and date, and retain in their loan file before processing a loan for faster processing. The information included in these forms are then compiled into a data set and are provided publicly pursuant to the Freedom of Information Act (FOIA). This loan origination data set includes basic information about the participants (i.e., the identity of the borrower and the lender and their addresses, city, zip code, and industry), nonpricing terms (i.e., loan volume, guarantee amount, or approval date), pricing term (i.e., loan spread plus base rate), ex-post loan performance (such as the total loan balance that has been charged off), and other administrative details such as the

[2] The following reports contain detailed information on the eligibility requirements: "Small Business Administration 7(a) loan guaranty program" by the Congressional Research Service; "Bankers' guide to the SBA 7(a) loan guaranty program" by the Office of Comptroller of the Currency Community Affairs Department; "Lender and development company loan programs. SOP 50 10 5(H)" by the SBA.

**Table 1**
Summary statistics.

*Notes:* This table shows summary statistics for the main analysis variables. The first two columns report the mean and the standard deviation, and the third to fifth columns report the 25th, median, and the 75th percentile, respectively. Panel A reports summary statistics for full sample, and panel B reports statistics for the sample of loans used in the notch estimation (loan size between $75,000 and $225,000). Loan amount is the size of a given loan in the sample. Reimbursement rate refers to the SBA determined reimbursement rate pooling across all years in the sample (2008–2017). Reimbursed amount is the guaranteed portion of the loan balance. Interest rate is the total interest rate (base plus spread) at the time of loan origination. Maturity is the length of loan terms, and charge-off amount is the total loan balance charged off, including the guaranteed and nonguaranteed portion of loan. Expected guarantee subsidy ($\Gamma$) is the predicted guarantee amount as a share of loan principal net of one-time and yearly fees. A full description of the methodology we use to estimate $\Gamma$ can be found in Appendix A.2. Loans per firm-lender pair reports the number of loans that a given firm borrows from the same lender in the same year. The excess mass reports an estimate of the amount of excess mass ($\hat{B}$) at the 150k notch, which we measure as the difference between the observed and counterfactual bin counts in the excluded region at and to the left of the notch. The estimate is reported as the share of bunching relative to the total number of loans in the estimation range. Excess mass is only reported in panel B, as it is estimated using the notch sample only. Source: SBA.

| Outcome | Mean | Std. dev. | $25^{th}$ Pctile. | Median | $75^{th}$ Pctile. |
|---|---|---|---|---|---|
| | | | A. Full sample | | |
| Loan amount ($) | 746,107 | 826,485 | 215,000 | 460,000 | 950,000 |
| Reimbursement rate | 80 | 6 | 75 | 75 | 85 |
| Reimbursed amount ($) | 574,195 | 626,519 | 168,750 | 356,400 | 735,000 |
| Interest rate (%) | 5.73 | 0.74 | 5.25 | 5.96 | 6.00 |
| Maturity (in years) | 15 | 8 | 10 | 10 | 25 |
| Charge-off amount ($) | 11,706 | 85,383 | 0 | 0 | 0 |
| Expected guarantee subsidy ($\Gamma$) | 0.02 | 0.04 | 0 | 0.02 | 0.05 |
| Loans per firm-lender pair | 1.05 | 0.27 | 1.00 | 1.00 | 1.00 |
| Observations | 199,013 | 199,013 | 199,013 | 199,013 | 199,013 |
| | | | B. Sample for notch estimation | | |
| Loan amount ($) | 147,359 | 41,330 | 112,000 | 150,000 | 180,000 |
| Reimbursement rate | 84 | 5 | 85 | 85 | 90 |
| Reimbursed amount ($) | 120,575 | 31,354 | 93,750 | 127,500 | 141,110 |
| Interest rate (%) | 6 | 1 | 6 | 6 | 6 |
| Maturity (in years) | 10 | 5 | 7 | 10 | 10 |
| Charge-off amount ($) | 6,221 | 26,704 | 0 | 0 | 0 |
| Expected guarantee subsidy ($\Gamma$) | 0.06 | 0.03 | 0.03 | 0.06 | 0.07 |
| Share of excess mass | 0.08 | 0.06 | 0.04 | 0.05 | 0.16 |
| Loans per firm-lender pair | 1.03 | 0.27 | 1.00 | 1.00 | 1.00 |
| Observations | 41,460 | 41,460 | 41,460 | 41,460 | 41,460 |

delegation status of the lender and the SBA district office that processed the loans.

For our analysis, we only consider loans originated over the last decade—2008 to 2017— under the SBA 7(a) program. We exclude SBA 7(a) Express loans and drop 22 loans that appear to contain data errors (i.e., loans for which the guaranteed share is greater than a hundred percent of the amount originated). Under these restrictions, the sample covers 199,013 loans originated by 3,066 lenders to 177,049 borrowers. Table 1 presents summary statistics for the main analysis variables.

The median SBA loan size is $460,000, and the guaranteed amount is $356,400. The median loan maturity and interest rate at the time of origination are ten years and 6%, respectively. Since the median prime rate is 3.25% in our sample, the maturity and interest rates are consistent with the SBA's maximum interest rate rule. Loans with a maturity of over seven years and amount greater than $50,000 can carry a maximum rate of 2.75% over the prime rate. The median charge-off amount is zero, while the mean is $11,706, indicating that the share of loans that are eventually charged off is small. Panel B of Table 1 reports the same statistics for subsample of loans used for notch estimation, where we restrict the loan size to be between $75,000 and $225,000. We restrict to the left of the threshold to loans above $75,000 to avoid the excluded region from a second interest rate notch. Loans below $50,000 carry a higher interest rate cap, which can additionally change lender incentives and lead to bunching. We take an equal range to the right of the $150,000 threshold to arrive at the upper bound, $225,000. Once we apply this restriction, we include 41,460 loans in the main analysis sample.

While the distribution is relatively similar to that in other papers using SBA data, such as Brown and Earle (2017), we only include 7(a) loans between 2008 and 2017. The difference in means relative to Brown and Earle (2017) comes from the fact that they include 504 loans that are up to $5.5 million, whereas we only examine loans below $350,000 in our main analysis sample. For certain heterogeneity analysis, we also link our main data to the Federal Deposit Insurance Corporation (FDIC) Statistics on Depository Institutions Data. This data set and sample construction is discussed in Appendix Section A.3. Additional robustness checks vary the main analysis sample to include some loans from the sample shown in panel A.

We use these data to estimate private lenders' responsiveness to federal loan guarantees. It is important to note that lenders cannot manipulate the lending structure by issuing multiple guaranteed loans to the same borrower. As discussed in the institutional details section, the SBA prohibits lenders from originating loans with a "piggyback" structure where multiple loans are issued to the same borrower at the same time, and the guaranteed loan is secured with a junior lien position. While this

policy does not prevent lenders from having a shared lien position with the SBA loans (i.e., Pari Passu), we confirm in our data that more than 99% of the borrowers receive only one loan from the same lender at the same time. As reported in Table 1, the average number of loans a given borrower receives from the same lender and year is one. The data also suggest that lenders are not "evergreening" loans–only (0.03%) of loans are categorized as "revolving" debt–and we remove these loans from the estimation sample. The institutional features of the SBA 7(a) program allows us to conduct a notch estimation for studying the impact of federal loan guarantees on credit supply.

## 3. Model and identification strategy

We model entrepreneurs as borrowing $D$ at interest rate $R$ from banks to fund their projects. Their projects are characterized by a productivity type that determines output and therefore the probability of success of the project. An entrepreneur's type is drawn from a distribution $F(r, n)$, characterized by the average type $r$ and variance $n$. While $r$ and $n$ are known to both borrower and bank, the realized type $\underline{r}$ is revealed only after the loan is made and the project is attempted.

Once a project's output is realized, borrowers decide whether to repay the loan to the bank. The borrower pays nothing in default and pays $D(1+R)$ otherwise. We assume that the borrower pays back as long as the realized output $\underline{r}$ is greater than the amount owed to the bank. Thus, the lender's payoffs are

$$\Pi = \begin{cases} -D & \text{if} \quad \underline{r} < D(1+R) \\ D\bar{R} & \text{if} \quad \underline{r} > D(1+R). \end{cases} \tag{1}$$

The lender loses the capital lent $D$ if the borrower defaults and gains $D\bar{R}$ if the borrower repays.

Lenders have market power but are restricted to charge a regulated interest rate of $\bar{R}$, which is consistent with interest rate caps in SBA programs. They decide how much capital $D$ to lend to a borrower by maximizing the expected profits:

$$\begin{aligned} E[\Pi] &= \int_{D(1+\bar{R})} D \cdot \bar{R} \cdot f(r)dr - \int^{D(1+\bar{R})} D \cdot f(r)dr \\ &= D \cdot \bar{R} \cdot Pr(r > D(1+\bar{R})) - D \cdot Pr(r < D(1+\bar{R})). \end{aligned} \tag{2}$$

The first term is positive and represents revenue made from a repaid loan. While the mechanical revenue, $D \cdot \bar{R}$, is increasing in loan size, the probability of repayment, $Pr(r > D(1+\bar{R}))$, is decreasing. This term is concave in $D$ so that it is equal to zero when $D$ is zero or infinite and is otherwise positive. The second term represents the expected costs to the lender from borrower default. The probability of default is given by $Pr(r < D(1+\bar{R}))$, which is increasing in loan size. Thus $-D \cdot Pr(r < D(1+\bar{R}))$ is negative, convex, and increasing in $D$.

We remain agnostic about the exact distribution of $r$, and write the probability of default as $\pi(D, \bar{R})$, an increasing function of $D$. Lender profits are:

$$E[\Pi] = D \cdot \bar{R} \cdot (1 - \pi(D, \bar{R})) - D \cdot \pi(D, \bar{R}). \tag{3}$$

Given the tradeoff between increased revenue and a higher default probability, lenders choose the loan size that maximizes their expected profits. Optimal loan size is implicitly a function of $\pi(\cdot)$ and satisfies the first-order equation:

$$D^* = \frac{\bar{R}}{\pi'(D^*, \bar{R}) \cdot (1+\bar{R})} - \frac{\pi(D^*, \bar{R})}{\pi'(D^*, \bar{R})}. \tag{4}$$

We focus only on the loans for which a positive $D^*$ exists given the distribution of realized output and the set interest rate, $\bar{R}$. Note that the optimal loan size will depend on the interest rate $\bar{R}$ as well as the mean and variance of realized productivity, which determine the shape of $\pi(D, \bar{R})$. All else equal, a borrower with a higher mean probability of default or higher variance will have a lower optimal loan size.

### 3.1. Lender's problem with a loan guarantee

We now analyze what happens to loan size when the lender receives an indirect loan guarantee. There are two key components of the federal loan guarantee program: a reimbursement rate and a fee. If a bank makes a loan that is ultimately charged off, the government will reimburse $\gamma\%$ of the losses. In return, the bank pays a certain fee equal to $\sigma$ percent of the loan principal to the government. Given a charge-off probability, $\pi(D, \bar{R})$, the total expected subsidy $S$ provided by the government on loan amount $D$ is given by

$$S = \gamma \cdot \pi(D, \bar{R}) \cdot D - \sigma \cdot D = D \cdot \Gamma. \tag{5}$$

where the net generosity of the guarantee per unit of lending is given by $\Gamma = \gamma \cdot \pi(D, \bar{R}) - \sigma$. We assume that banks are risk-neutral so that a change in the reimbursement rate is isomorphic to a change in the fee.

The guarantee does not change the borrower's behavior since it is targeted toward and given only to the lender. Indeed, the guarantee is a contract between the lender and the government and hence should not directly affect borrowers other than through lender behavior. The lender's payoffs are now

$$\Pi = \begin{cases} \gamma D - (1+\sigma)D & \text{if} \quad \underline{r} < D(1+R) \\ \bar{R}D - \sigma D & \text{if} \quad \underline{r} > D(1+R). \end{cases} \tag{6}$$

and the expected profits are

$$\begin{aligned} E[\Pi] = D \cdot (\bar{R} - \sigma) \cdot (1 - \pi(D, \bar{R})) \\ - (1 + \sigma - \gamma) \cdot D \cdot \pi(D, \bar{R}). \end{aligned} \tag{7}$$

The guarantee decreases marginal revenue since it requires paying a fixed percentage fee, $\sigma$. However, in the case of a subsidy, the reimbursement component also decreases the marginal cost of lending from $\pi'(D, \bar{R})$ to $(1 + \sigma - \gamma) \cdot \pi'(D, \bar{R})$. We analyze what happens when the guarantee is made more generous using this profit function–specifically, what happens to profit and loan size when $\gamma$ increases holding all else constant?

We focus on positive subsidy guarantees such that $\sigma = 0$, $\gamma > 0$, and $\Gamma = \gamma \cdot \pi(D)$. Taking the derivative of expected profit with respect to loan size gives us a new

formula for $D^*$ that relies on the guarantee generosity:

$$D^* = \frac{\bar{R}}{(1+\bar{R})\cdot\pi'(D^*,\bar{R}) - \Gamma\frac{\pi'(D^*,\bar{R})}{\pi(D^*,\bar{R})}} - \underbrace{\frac{\pi(D^*,\bar{R})}{\pi'(D^*,\bar{R})}}_{\text{Default effect}}. \tag{8}$$

This expression shows that the elasticity of loan size to the guarantee depends not only on the generosity of $\Gamma$ but also on the size and shape of the default function. While a more generous guarantee decreases the costs of default borne by the lender–inducing lenders to increase the loan supply, $D^*$–a larger loan carries a higher probability of default. The magnitude of the elasticity of loan size to the guarantee is therefore inversely related to the local slope of the default function.

The local slope of the default function is determined by the productivity type distribution, $F(r, n)$. Specifically, as the variance of the productivity type increases, an equal sized change in $D$ will cause a smaller change in the default probability. Thus, an increase in $n$ flattens the slope of the default function and leads to higher lending supply elasticity with respect to $\Gamma$.

The top panel of Fig. A.1 simulates how loan size responds to a varying type of $n$. The change in the loan size is positively related to the variance of productivity type distribution, illustrating that the increase in the variance of expected returns leads to higher lending response with respect to $\Gamma$ through a flattening of the slope of the default function. The bottom panel simulates changes in $D^*$ as $\Gamma$ increases for a high and low variance distribution of expected returns. Again, the guarantee has a larger loan size effect for the high variance distribution, which becomes amplified as the subsidy increases in generosity.

### 3.2. Impact of guarantee subsidy on lender profits versus additional lending

An increase in guarantee generosity is costly for the government. To what extent does this spending simply subsidize lenders, and how does the subsidy versus loan creation effect depend on the loan size elasticity? Given that $D^*$ is implicitly a function of $\Gamma$, we rewrite lender profit as $E[\Pi] = D^*(\Gamma)\cdot(\bar{R}\cdot(1-\pi(D,\bar{R})) - \pi(D,\bar{R}) + \Gamma)$ and take the derivative with respect to $\Gamma$:

$$\frac{\partial E[\Pi]}{\partial \Gamma} = \underbrace{D'(\Gamma)\cdot(\bar{R}\cdot(1-\pi(D,\bar{R})) - \pi(D,\bar{R}) + \Gamma) + D}_{\text{Increased revenue from larger loan}} - \underbrace{D(\Gamma)\cdot(1+\bar{R})\cdot\pi'(D,\bar{R})\cdot D'(\Gamma)}_{\text{Decreased prob. of repayment}} \tag{9}$$

While complex, this derivative shows that profits change due to both loan size adjustment and the mechanical decrease in expected costs. Therefore, the extent to which the guarantee acts as a lender subsidy relies on the responsiveness of loan size to the guarantee rate, $D'(\Gamma)$.

Recall that the expected total cost of the guarantee subsidy is $S = D\cdot\Gamma$. If loan size is completely inelastic, the change in profits will be exactly equal to the change in costs, and the guarantee will act as a pure subsidy to lenders. As loan size becomes more responsive to the guarantee, the expected costs of the guarantee and net-of-guarantee losses for the lender increase. While the loan size increases more dramatically, less of the guarantee subsidy is retained by the lender. Fig. 1 illustrates this logic. The top panel shows that the fraction of the guarantee subsidy that goes to the lender declines as $D'(\Gamma)$ increases, while the bottom panel shows that the lending supply expands with $D'(\Gamma)$.

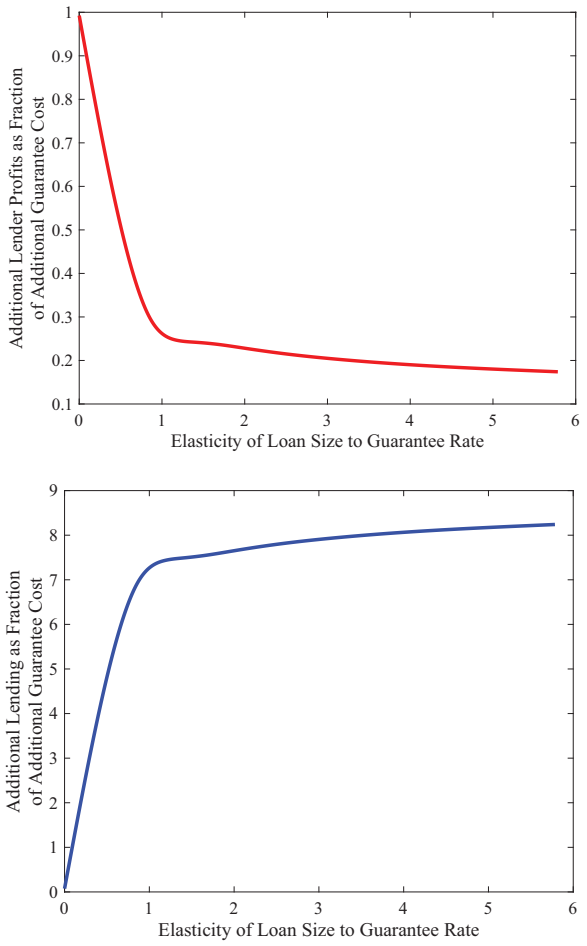



**Fig. 1.** Changes in bank profit and lending by elasticity.
*Notes:* Here we simulate what happens to bank profits and additional dollar lending when the reimbursement rate changes from $\gamma = 0.5$ to 0.8. The top figure plots the change in bank profit retained by the lender as a fraction of the change in the guarantee subsidy cost, $S$. The figure illustrates that if the elasticity of loan size to guarantee rate is inelastic, the share of guarantee subsidy retained by the lender is high and this share declines as the elasticity increases. The bottom figure plots the change in additional lending as a fraction of the change in the guarantee subsidy cost over lending elasticities. As the lending supply becomes more responsive to the guarantee, the expected costs of the guarantee and the net-of-guarantee losses for the lender increase. Therefore, loan size increases as the elasticity increases, while the guarantee subsidy retained by the lender decreases.

### 3.3. Identification strategy

Our identification strategy and the interpretation of our estimated elasticity relate closely to the curvature of the lenders' profit function modeled in Section 3.1. Specifically,

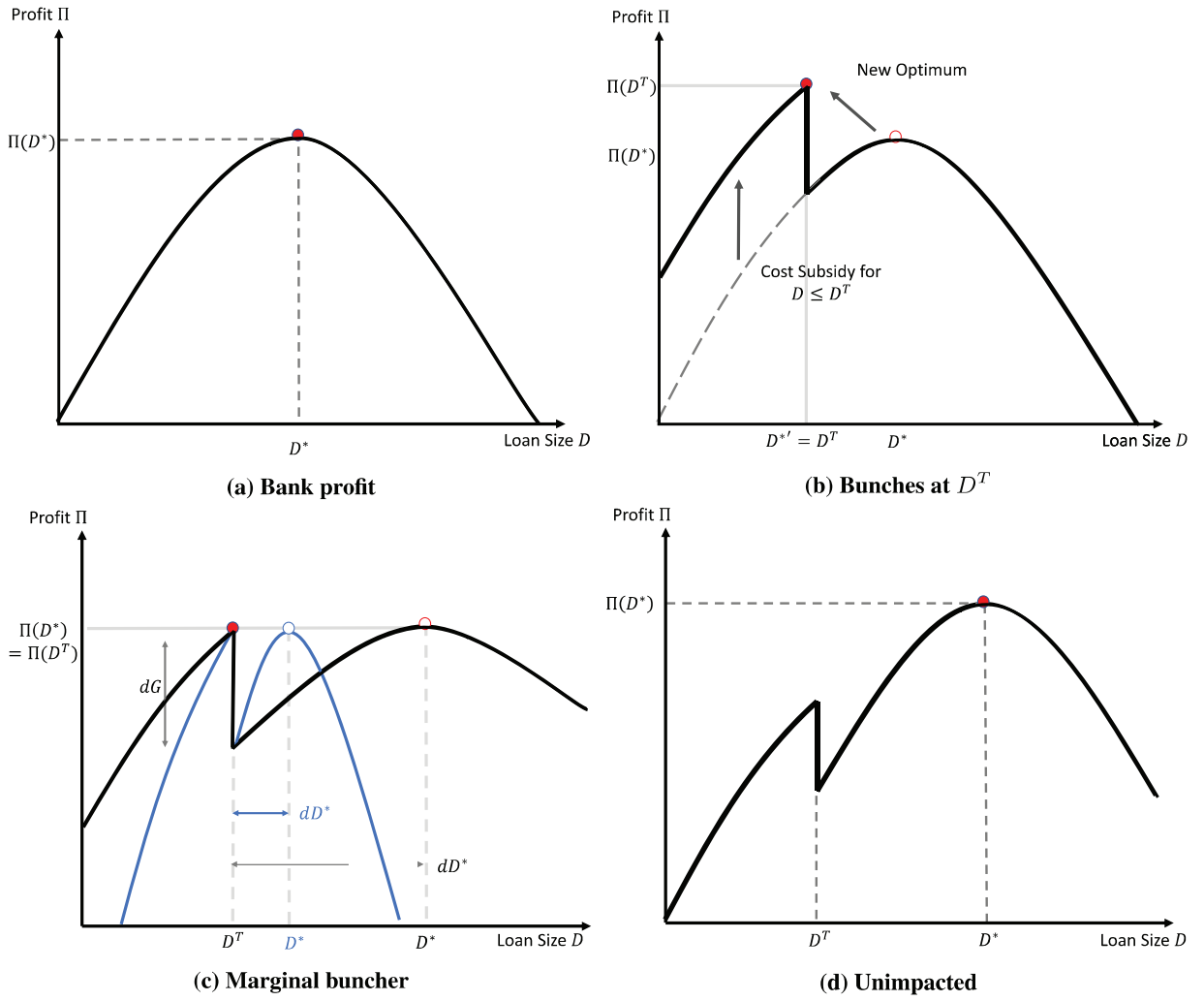



**Fig. 2.** Bank profit as a function of $D$ for a given observable type $r$.
*Notes:* This figure illustrates the shape and the slope of a bank's profit as a function of loan size $D$ for a given productivity type $r$. Panel a illustrates a concave profit function in the absence of a notch where the location of the global optimum $D^*$ is indicated by a red dot. Panel b shows a new profit function when a notch point is introduced at $D^T$. Panel c shows a profit function for a marginal buncher as well as an additional profit function with a steeper slope (in blue). The shape of the profit function is determined by the distribution of realized productivity types, $F(r, n)$ and the interest rate $\bar{R}$. Panel d shows a profit function of an unimpacted borrower. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

since lenders' profit functions are concave, we assume that there is a global optimum of the amount of capital $D_i^*$ that the bank should lend to each borrower with mean productivity type, $r_i$. This optimum is shown in panel a of Fig. 2; the red dot indicates the point where a lender maximizes profit for a given borrower type.

We assume that there is a distribution of mean productivity types in the population, and thus the optimal amount of capital varies by the expected type. Therefore, even in the absence of the guarantee notch, this leads to wide variation in the amount of capital lent to the borrowers. Fig. A.2 illustrates this point–the observed loan distribution is wide, even in the placebo years when the guarantee notches were eliminated, indicating that the heterogeneity in loan size is driven by the underlying productivity types.

In our setting, we observe a loan size-specific guarantee subsidy that creates a discontinuity in the profit function with respect to $D$. A more generous guarantee applies to all loans below a specific loan size threshold, $D^T$. All else constant, this shifts the bank's profit function upwards in this region. As shown in the panel b of Fig. 2, the notch creates a new optimum for a certain subset of productivity types. In particular, for some borrowers that normally would be optimally located to the right of the notch, the notch will distort the distribution of observed loans, as it will now be profit maximizing for the bank to offer $D_i = D^T$. It is important to note that this will only impact the placement of loans that were previously located to the right of the notch. If the optimum was previously to the left of the notch, the guarantee will change the level of profit received by the bank but not the location of $D_i^*$.

Whether a given productivity type is affected by the notch is determined by how profit changes between $D_i^*$ and $D^T$. Fig. 2 illustrates this point. For the borrower in panel b, the notch creates a new optimum. However, panel d shows that for the borrowers with an original optimum $D_i^*$ further to the right away from $D^T$ will be less likely to be relocated to the notch. This is because there is a smaller difference in the profit at $D_i^*$ and $D^T$ as $D_i^*$ increases. Finally, panel c shows the borrowers that we refer to as the "marginal buncher," or those that the bank is indifferent between giving a loan at either $D_i^*$ or $D^T$.

Our estimation strategy, which recovers the local slope of the profit function, relies on identifying the marginal buncher. We do this by comparing the observed distorted and counterfactual undistorted loan distributions. We identify the point to the right of the notch where the observed loan distribution is no longer distorted or impacted by the notch. This corresponds to the location of the marginal buncher. We define the distance between the undistorted optimal location of the marginal buncher and $D^T$ as $\Delta D$, and it is the key empirical determinant of our reduced-form elasticity. The method assumes homogeneity in profit function across types $r_i$. In our current setup, this implies that the riskiness of the realized draw $n$ does not vary with mean expected returns, $r$.

The location of the marginal buncher, and hence the measured elasticity, depends on the curvature of the profit function. Panel c of Fig. 2 plots both a very steep profit function (in blue) and flat profit function (in black) that both face a guarantee notch with the same size and location. It denotes the location of the marginal buncher in each case. The reduced-form elasticity that we estimate maps approximately to the inverse of the slope near and to the left of the optimum. As discussed above, this underlying curvature is determined by the underlying distribution of realized types $F(r, n)$ and the interest rate $\bar{R}$.

## 4. Empirical approach

As explained in Section 3.3, we identify and estimate the elasticity of lending to a change in the guarantee rate using the discrete change in the level of the guarantee rate in the SBA 7(a) lending program. The notch point created by the change in the guarantee rates creates incentives for lenders to shift loans below the guarantee notch point. If lending is elastic to the guarantee rate, lenders will be more likely to shift loans to a point below the notch where a loan carries a higher guarantee rate, whereas if lending is inelastic, lenders will not alter their behavior. Specifically, an elastic response will lead to "bunching" at the notch point, with excess mass below the notch point where guarantee rates are higher and missing mass above the notch point where guarantee rates are lower.

A bunching approach uses the excess mass at the threshold to estimate an implied lending response to the change in the guarantee rate and provides nonparametric estimates of the elasticity of credit supply. Recent papers employing bunching estimators include Kleven (2016), Best and Kleven (2018), DeFusco and Paciorek (2017), Saez (2010), Kleven and Waseem (2013). The method is related to, but distinct from a regression discontinuity approach. Regression discontinuity design exploits notched incentives when there is no manipulation of an assignment variable. In a bunching design, the manipulation of the assignment variable is used to identify the parameter of interest (see Kleven, 2016 for a general overview of bunching estimators). In the subsequent analysis, we closely follow the methodology outlined in Kleven and Waseem (2013).

To implement the approach, we first recall that a bank $i$ decides how much to lend, $D_{ij}$, to entrepreneur $j$ using the objective function that maximizes returns in $D_{ij}$:

$$\max_{D_{ij}} D_{ij} \cdot (\bar{R} \cdot (1 - \pi(D_{ij}, \bar{R})) - \pi(D_{ij}, \bar{R}) + \Gamma_{ij}). \tag{10}$$

We calculate $\Gamma_{ij}$ as the observed ex-post return on a loan, net of realized charge-offs, guarantee fee payments, and guarantee reimbursements. We use our loan-level data to first model an indicator for loan default as a function of loan size. We multiply the predicted default probabilities $(\pi)$ by the guaranteed reimbursement rate $(\gamma)$ to find the expected reimbursement rate on a given loan–this implicitly assumes a hundred percent charge-off rate on defaulted loans. We then subtract the loan fees $(\sigma)$ paid to the SBA, which are expressed as a percentage of loan principal. This provides the net subsidy rate provided to banks by the SBA, the empirical analogue to $\Gamma = (\gamma \cdot \pi - \sigma)$ in Section 3.1. A full description of the methodology we use to estimate $\Gamma$ can be found in Appendix A.2.

Empirically, the default probability varies little across the threshold, whereas $\gamma$ and $\sigma$ vary significantly. We make the assumption that banks are risk neutral, which means that lenders treat a change in the reimbursement rate equivalently to a change in the fee. This generates a discrete drop in the return the bank makes on lending right above the threshold. Specifically,

$$\Gamma(D_{ij}) = \begin{cases} \Gamma, & \text{if } D_{ij} \leq D^T \\ \Gamma - \Delta\Gamma, & \text{otherwise.} \end{cases}$$

In the absence of a notch, we assume there would have been a smooth distribution of loans made that would satisfy the banks' first-order condition, conditional on and mapping directly to a smooth underlying distribution of loan demand, $n_j$. The notch, however, creates a region directly above the threshold for a subset of loans where marginal revenue is strictly lower than the marginal cost. The marginal bunching loan is made at the point $D^T + \Delta D$ where the bank is indifferent between making a smaller loan under the more generous guarantee and making a larger loan under the less generous guarantee:

$$\begin{aligned} &D^T \cdot (\bar{R} \cdot (1 - \pi(D^T, \bar{R})) - \pi(D^T, \bar{R}) + \Gamma) \\ &= (D^T + \Delta D) \cdot (\bar{R} \cdot (1 - \pi(D^T + \Delta D, \bar{R})) \\ &\quad - \pi(D^T + \Delta D, \bar{R}) + (\Gamma - \Delta\Gamma)). \end{aligned} \tag{11}$$

Therefore, $\Delta D$ captures the reduction in dollars lent in response to the change in the guarantee rate for this marginal buncher, and it is the key empirical parameter needed to calculate the elasticity of lending. The substantial excess mass we observe in the data at the point $D_{ij} = D^T$ comes from this region of strictly dominated lending for the bank $(D^T, D^T + \Delta D)$ directly above the notch point. This approach allows us to map the

amount of excess mass to the loan response $\Delta D$ using the bunching methodology we discuss below in Section 4.1.

Within the dominated region, the bank can always increase its return by making smaller loans under the higher guarantee rate, $\Gamma$. As discussed in Section 3.3, the size of the dominated region (and therefore the reduced-form elasticity of lending with respect to the guarantee rate) relates to the slope of the default function $\pi(D)$–if a small change in $D$ generates a sharp increase in costs, there will be a small dominated region and a small elasticity of lending. If a change in $D$ has little impact on costs, then there will be a larger dominated region, more bunching at the threshold, and a larger elasticity of lending with respect to the guarantee rate. The analysis treats the guarantee parameter as exogenously set by the SBA.

### 4.1. Bunching methodology

This section describes the estimation methodology in detail. Our objective is to estimate the reduced-form lending elasticity with respect to the guarantee generosity, or the percentage change in dollars lent that results from a corresponding percentage change in the guarantee generosity:

$$\varepsilon_{D,\Gamma} \equiv \frac{\Delta D}{D^T} \times \frac{(1+\Gamma^T)}{\Delta\Gamma}. \tag{12}$$

Here $\Delta\Gamma$ is the change in the marginal guaranteed return faced by the bank. We estimate the elasticity by noting that a notch in the marginal guarantee rate allows us to approximate the implicit marginal guarantee rate, $\Gamma^T$, created by the notch $\Gamma^T \approx \Gamma + \Delta\Gamma \frac{D^T}{\Delta D}$. We can then write the reduced form elasticity as

$$\varepsilon_{D,\Gamma} \approx \left(\frac{\Delta D}{D^T}\right)^2 \times \frac{(1+\Gamma)}{\Delta\Gamma}. \tag{13}$$

The validity of the bunching estimate relies on three keys assumptions: first, the counterfactual distribution would be smooth in the absence of a notch; second, bunchers come from a continuous set such that there exists a well defined marginal buncher; third, optimization frictions are locally constant. Interpreting this as purely the effect of a change in the guarantee rate requires a fourth assumption that contract terms do not change at the notch point due to the presence of a guarantee.

The first assumption rules out that other factors are changing at the threshold, which might bias our estimates. The assumption effectively means that there are no other policies at the threshold that would induce borrowers to move to the notch point. To our knowledge, there are no other relevant contract parameters, and we confirm this empirically for observable contract parameters in the data. This assumption also captures extensive margin responses and implies that locally borrowers move to the notch rather than choosing not to embank on projects.

While the second assumption is technical and fairly weak, the third assumption is stronger. The assumption that optimization frictions are locally constant allows the use of the dominated region to the right of the notch to identify behavioral responses and parameters of interest. This assumption requires that the mass of the set of movers equals the total area under a counterfactual on the other side of the notch point.

The assumption that contract terms do not change at the notch point due to the presence of a guarantee is likely to hold in our context due to the particular institutional details. The main parameter that lenders might vary in response to the guarantee is price. Empirically, we observe that interest rates trend smoothly across the notch. This is likely due to the presence of interest rate caps–the vast majority of lenders price right at the cap.

It is important to note that there are a wide variety of current and historical government guarantee programs, ranging from the mortgage and student loan markets to the small business loans that we study. Our estimates are for small business loans between $75,000 and $225,000. It is possible that outside the range, lending supply could be more or less responsive, and it is also possible that effects are different in other loan markets.

We obtain the parameters for elasticity estimation from the SBA data. The threshold $D^T$ is $150,000 for the years in our sample. We calculate $(1+\Gamma)$ as the observed ex-post return on a loan, net of realized charge-offs, guarantee fee payments, and guarantee reimbursements. As noted earlier, interest rates and ex-post charge-off rates trend smoothly through the threshold. Therefore, all systematic variation in returns come from changes in the generosity of the guarantee contract at the threshold. Over our time period, loans less than or equal to $150,000 had lower guarantee fees and higher guarantee reimbursement rates than loans to the right of the threshold. Given that the generosity varies over time, we estimate the excess mass and elasticity separately by year. Note that while the effective generosity of the guarantees vary over time, the charge-offs are low and stable over time. Fig. A.3 shows that the three-year cohort default rates are relatively stable except during the Great Recession. Therefore our elasticity measure is identified through the variation in the guarantee rates rather than time-varying default rates.

To calculate $\Delta D$ empirically, we must locate the counterfactual loan amount provided to the marginal buncher. This occurs at the point where the excess mass at the threshold is equal to the missing mass to the right of the threshold. To measure the excess and missing mass, we estimate the counterfactual loan distribution that would have occurred in the absence of a notch by fitting a polynomial of degree six with a vector of round number dummies for multiples of 1, 5, 10, 25, and 50 thousand and excluding a region at and to the right of the threshold:

$$N_j = \sum_{k=0}^{6} \beta_k (d_j)^k + \sum_{i=d_l}^{d_u} \delta_{ij} \mathbb{1}(d_j = i) + \sum_{n \in \{1k,5k,10k,25k,50k\}} \delta_n \mathbb{1}(d_j = n) + \eta_j. \tag{14}$$

where $N_j$ is the number of loans in bin $j$, $d_j$ is the loan amount midpoint of interval $j$, $\{d_l, d_u\}$ is the excluded region, $\delta_{ij}$ are dummies for bins for the excluded region, and $\delta_n$ are dummies for multiples of prominent round numbers. For estimation, we cut the data into $500 dollar bins and restrict the loan size to be between $75,000 and

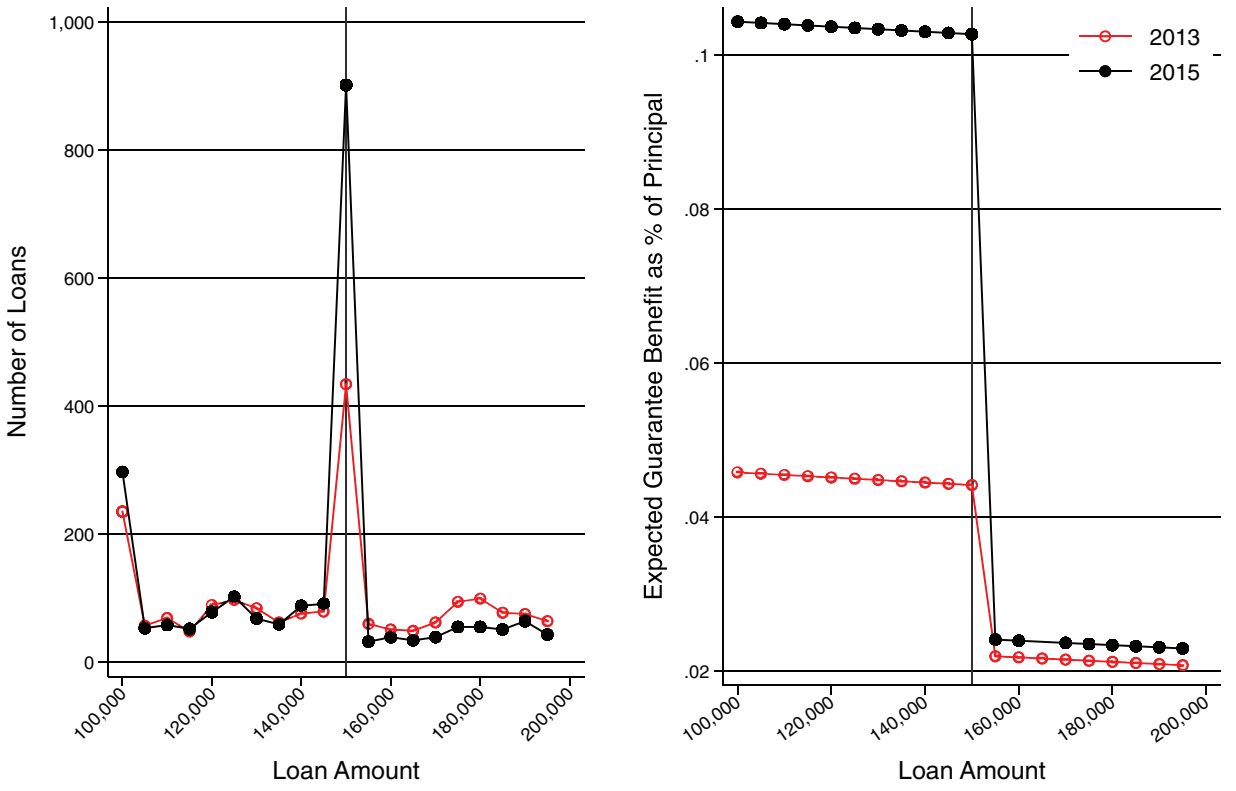



**Fig. 3.** Bunching at the guarantee notch in 2013 and 2015.
*Notes:* The left panel shows the number of loans made in discrete $5,000 bins made in 2013 (red) and 2015 (black). The right panel shows the change in the guarantee rate at the threshold in these two years. In 2015, when the change in the guarantee at the threshold was much larger than in 2013, there was substantially more excess mass at the threshold. Source: SBA. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$225,000 to limit the estimation range. For robustness, we repeat the estimation with bin sizes of $200, $1000, and $2000, polynomials of degree four, five, and seven, and for various ranges of loan samples; these results are shown in the appendix. While the results are very robust to the different bin and polynomial choices, they are sensitive to the inclusion of $50,000 within the range. Another interest-rate-related threshold exists at the $50,000 mark, which causes additional bunching, and therefore we exclude it from our estimation. The counterfactual distribution, $\hat{N}_j$, is estimated as the predicted values from Eq. (14) using the $\beta_k$ and the $\delta_n$ terms

$$\hat{N}_j = \sum_{k=0}^{6} \hat{\beta}_k (d_j)^k + \sum_{n \in \{1k,5k,10k,25k,50k\}} \hat{\delta}_n \mathbb{1}(d_j = n). \qquad (15)$$

Excess mass is defined as the difference between the observed and counterfactual bin counts between the lower limit of the excluded region ($d_l$) and the threshold, $\hat{B} = \sum_{j=d_l}^{D^T} (N_j - \hat{N}_j)$, whereas the missing mass, $\hat{M} = \sum_{j=D^T}^{d_u} (N_j - \hat{N}_j)$, is defined as the same bin counts but is in the range between the threshold and the upper limit of the excluded region ($d_u$).

While the lower limit $d_l$ is easily observable visually as the notch point, we do not observe a sharp valley to the right of the cutoff. This is common in bunching estimators (Kleven and Waseem, 2013). Thus, to identify the upper limit $d_u$, we follow an iterative procedure. We identify the upper limit (i.e., $d_u = D^T + \Delta D$) by requiring that the excess mass $\hat{B}$ be equal to the missing mass, $\hat{M}$.

The estimation procedure proceeds in four steps. First, the estimation begins with a starting value of $d_u$ right above $D^T$. Second, we calculate $(\hat{B} - \hat{M})$. The next step is to increase $d_u$ by a step size of $500 if $(\hat{B} - \hat{M} \neq 0)$. Finally, we repeat these steps until the result converges. We pool together all banks in our main estimation. However, to test whether the elasticity and bunching is driven by a specific bank, we have also repeated the estimation on a conditional distribution that controls for bank fixed effects. The bunching and elasticities are very similar.

We calculate standard errors for Eq. (13) using a non parametric bootstrap procedure in which we draw a large number of loan distributions following Chetty et al. (2011). We create new bins of loans by drawing randomly with replacement from the estimated vector of $\eta_j$ and by adding those to the estimated distribution implied by the coefficients $\{\hat{\beta}_k, \hat{d}_j, \hat{d}_u\}$ from Eq. (14). Finally, we apply the bunching estimator technique described above again to calculate a new estimate $\hat{\varepsilon}^b_{D,\Gamma}$. We repeat this procedure 1,000 times and define the standard error as the standard deviation of the distribution of $\hat{\varepsilon}^b_{D,\Gamma}$s created. As we observe the universe of SBA 7(a) loans over the years considered, the standard error represents error due

**Table 2**
Guarantees and fees by loan amount.

*Notes:* This table includes fees and guarantee rates for loans with maturities longer than 12 months. Fees are calculated as a percentage of the loan principal. The reimbursement rate is expressed as a percentage of charged-off principal. The net benefit combines the fees and reimbursement rate to measure the average expected generosity of the guarantee and is expressed as a percentage of the loan principal amount. This net benefit is calculated as the guaranteed reimbursement on expected losses minus guarantee fees. Loan amounts smaller than $150,000 refers to loans between $0-150,000. Loan amounts larger than $150,000 refers to loans between $150,000-700,000. Source: SBA.

| | Loan amount smaller than $150,000 | | | | Loan amount larger than $150,000 | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Fiscal year | Yearly fee | One time dee | Reimbursement rate | Net benefit | Yearly fee | One time fee | Reimbursement rate | Net benefit |
| 2008 | 0.55 | 2 | 85 | 4.6 | 0.55 | 3.42 | 75 | 2.6 |
| 2009 | 0.55 | 0 | 90 | 7.4 | 0.55 | 0 | 90 | 7.4 |
| 2010 | 0.55 | 0 | 90 | 7.4 | 0.55 | 0 | 90 | 7.4 |
| 2011 | 0.55 | 2 | 85 | 4.9 | 0.55 | 3.42 | 75 | 2.9 |
| 2012 | 0.55 | 2 | 85 | 4.6 | 0.55 | 3.42 | 75 | 2.7 |
| 2013 | 0.55 | 2 | 85 | 5.8 | 0.55 | 3.42 | 75 | 2.6 |
| 2014 | 0 | 0 | 85 | 10.5 | 0.52 | 3.42 | 75 | 2.9 |
| 2015 | 0 | 0 | 85 | 10.5 | 0.52 | 3.42 | 75 | 2.9 |
| 2016 | 0 | 0 | 85 | 9.6 | 0.47 | 3.42 | 75 | 2.9 |
| 2017 | 0.55 | 0 | 85 | 6.3 | 0.55 | 3.42 | 75 | 2.7 |

**Table 3**
Excess mass and elasticity estimates.

This table reports estimates of excess mass and the main elasticity estimates. The top panel shows placebo years (2009 and 2010) where there was no change in the reimbursement rate at the 150,000 threshold. The bottom panel shows years where a notch existed (2008, 2011–2017). Elasticity estimates are reported in the latter sample. For estimation, we restrict the loan sample with size between $75,000 to $225,000, use the step size of 500, and include round number dummies for multiples of 1,5, 10, 25, and 50 thousand. The polynomial used is denoted in the second column. The change in the guarantee rate ($\Delta\Gamma$) at the threshold for years in which a notch existed is computed as the weighted average of the average expected guarantee benefit as a percentage of the loan principal, where the weights correspond to the number of loans across years 2008, 2011–2017. Standard errors are reported in italics and are obtained by empirical bootstrap with 1000 repetitions of resampling the distribution of loans made. The bunching estimation routine is run at every bootstrap iteration until convergence. Source: SBA.

| Year | Polynomial | Excess mass | $\Delta$D | $\Delta\Gamma$ | Elasticity |
|---|---|---|---|---|---|
| | | | *A. Placebo years - no notch* | | |
| 2009–2010 | 5 | 67 | 21,000 | – | – |
| | | *(21.57)* | *(14,796)* | – | – |
| | 6 | 66 | 21,000 | – | – |
| | | *(41.36)* | *(14,285)* | – | – |
| | 7 | 0 | 9500 | – | – |
| | | *(16.76)* | *(13,131)* | – | – |
| | | | *B. Pooled years - with notch* | | |
| 2008, 2011–2017 | 5 | 4,744 | 66,500 | 0.038 | 4.519 |
| | | *(98.9)* | *(1,326)* | – | *(0.186)* |
| | 6 | 4,747 | 66,000 | 0.038 | 4.589 |
| | | *(44.38)* | *(2,806)* | – | *(0.395)* |
| | 7 | 4,745 | 70,500 | 0.038 | 5.235 |
| | | *(102.6)* | *(1,240)* | – | *(0.181)* |

to misspecification of the polynomial and the number of dummies included in the exclusion zone used in rather than representing sampling error.

Fig. 3 visually illustrates the variation that we use to identify the elasticity of credit supply to the loan guarantee. The figure shows the raw data in 2013, where the guarantee notch is relatively small, and again in 2015 when the guarantee notch is larger. Fig. 3 illustrates the striking contrast in bunching in 2013 when there was a small notch, and in 2015 when there was a large notch. The left panel shows the number of loans, in discrete $2,000 bins, while the right panel shows the total expected guarantee benefits. In 2015, where the total expected guarantee is comparatively higher, we see more bunching relative to 2013.

The bunching technique captures intensive margin responses. If banks reject applications simply because they are above the threshold, this would lead us to underestimate the credit supply response to the guarantee further away from the notch and to make our estimates more sensitive to the choice of polynomial. Since banks have considerable power when deciding how much to lend and could increase returns by reducing $D_{ij}$ rather than not lending at all, these extensive margin responses are unlikely in our setting. However, we still test the sensitivity of our estimates to the choice of parameters. Kleven and Waseem (2013) show that these extensive margin responses should only occur in a region far to the right of the notch, with the intensive margin response concentrated in the area directly next to the notch. They note that extensive margin bias will mainly enter via functional form misallocation, and therefore sensitivity analysis should be conducted with respect to the polynomial. We show in Table 3 that our results are robust to using

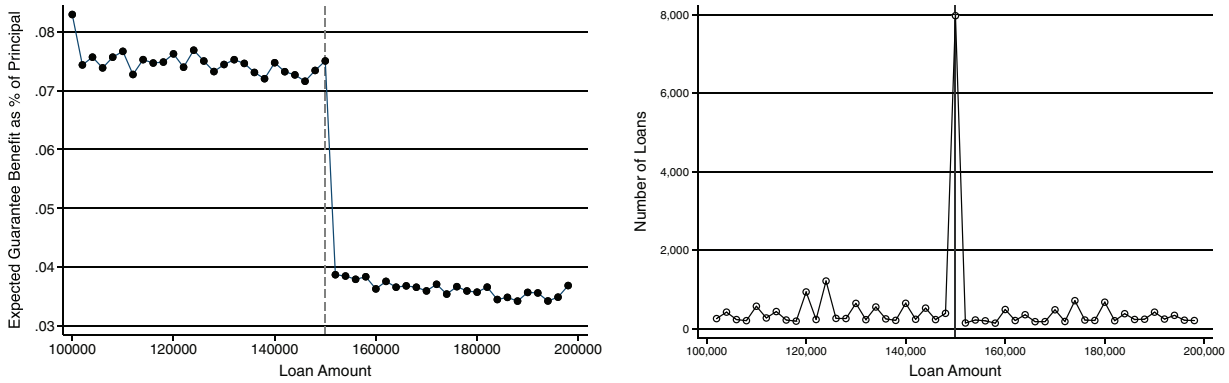



**Fig. 4.** Guarantees and fees by loan amount.
*Notes:* The left panel shows the average expected guarantee benefit as a percentage of the loan principal amount for discrete $2000 bins across the threshold. This net benefit is calculated as the guaranteed reimbursement on expected losses minus guarantee fees. The right panel shows the number of loans made in discrete $2000 bins across the threshold. The figures pool over all years 2008–2017. Source: SBA.

a range of polynomials, which suggests that extensive margin responses do not play a large role in our setting.

## 5. Main results

### 5.1. Visual evidence

We begin by showing the change in guarantees at the $150,000 threshold. The left panel of Fig. 4 shows average guarantees and fees by loan amounts as a percentage of the loan principal amount in $2000 bins across the threshold between 2008 and 2017. Consistent with the policy rule, the guarantee benefit jumps sharply across the threshold–loans below $150,000 receive a guarantee rate nearly twice as generous as loans above the threshold. Appendix Fig. A.4 breaks down the guarantee benefit by the average expected guarantee fees and reimbursement rate separately.

To determine whether the guarantee benefit notch affects lending volumes, we analyze the density of borrowing. The right panel of Fig. 4 shows bunching directly below the threshold. The figure shows the number of loans in $2000 bins across the threshold between 2008 and 2017. Visual evidence indicates that there are significantly more loans at the threshold relative to other points nearby, consistent with banks lending fewer dollars in response to a lower guarantee rate (i.e. moving borrowers to loan volumes below the notch).

Fig. 5 shows the observed and counterfactual density of loans. The solid line shows the observed number of loans, while the dashed line shows the counterfactual number of loans. The counterfactual is determined according to the method discussed in Section 4 and is estimated as specified in Eq. 15. Several patterns are immediately clear from Fig. 5. First, there are significantly more loans disbursed just at the threshold, which is consistent with guarantees affecting credit supply. Second, there is also missing mass to the right of the guarantee notch. In other words, the counterfactual distribution is higher than the observed distribution. Third, the observed number of loans is lower to the right of the threshold. Finally, there is significant round number bunching, which is captured by our modeling procedure.

The presence of two placebo years in 2009 and 2010 in the middle of our sample period, when no notch existed, provides a direct test of the first assumption that the counterfactual distribution would be smooth in the absence of a notch. Fig. A.5 shows that the bunching disappears completely in these years and suggests that there are no other unobserved factors generating bunching at the threshold. These years also allow us to test the fit of our estimated counterfactual not only at the notch but also across the rest of the loan distribution. The observed and estimated distributions in Fig. A.5 are almost identical, which indicates that our counterfactual specification accounts well for the round number bunching in the distribution.

### 5.2. Elasticity estimates

Table 3 formalizes and scales the bunching noted above relative to the change in the size of the guarantee and presents estimates of $\varepsilon_{D,\Gamma}$ as described in Section 4. The first column shows the degree of the polynomial used to estimate the counterfactual distribution- we vary this to test sensitivity to the parameter choices and to gauge whether extensive margin responses are playing a large role. The second column shows the estimated excess mass, $\hat{B}$, in terms of the number of loans. The third column shows estimates of $\Delta D$, the distance of the marginal buncher in dollar terms from the threshold. The fourth column presents $\Delta\Gamma$, the change in the generosity of the guarantee rate at the notch. Over the years in our sample, $\Delta\Gamma$ varied between 0 and 0.078. For this estimate, we take a weighted average of $\Delta\Gamma$ in non zero years to pool across years; in the appendix we also list estimates by year. The final column shows estimates of $\varepsilon_{D,\Gamma}$, the elasticity of dollars of loans made with respect to the guarantee rate.

The first panel show estimates from placebo years when the notch was eliminated as part of the ARRA stimulus of 2009. Reassuringly, we see very little excess mass when loan guarantees are identical across the notch. This assuages potential concerns that other factors could be changing across the threshold and is discussed further in the next section. Note that we cannot compute elasticity estimates in 2009 and 2010, as there is no variation in the notch.

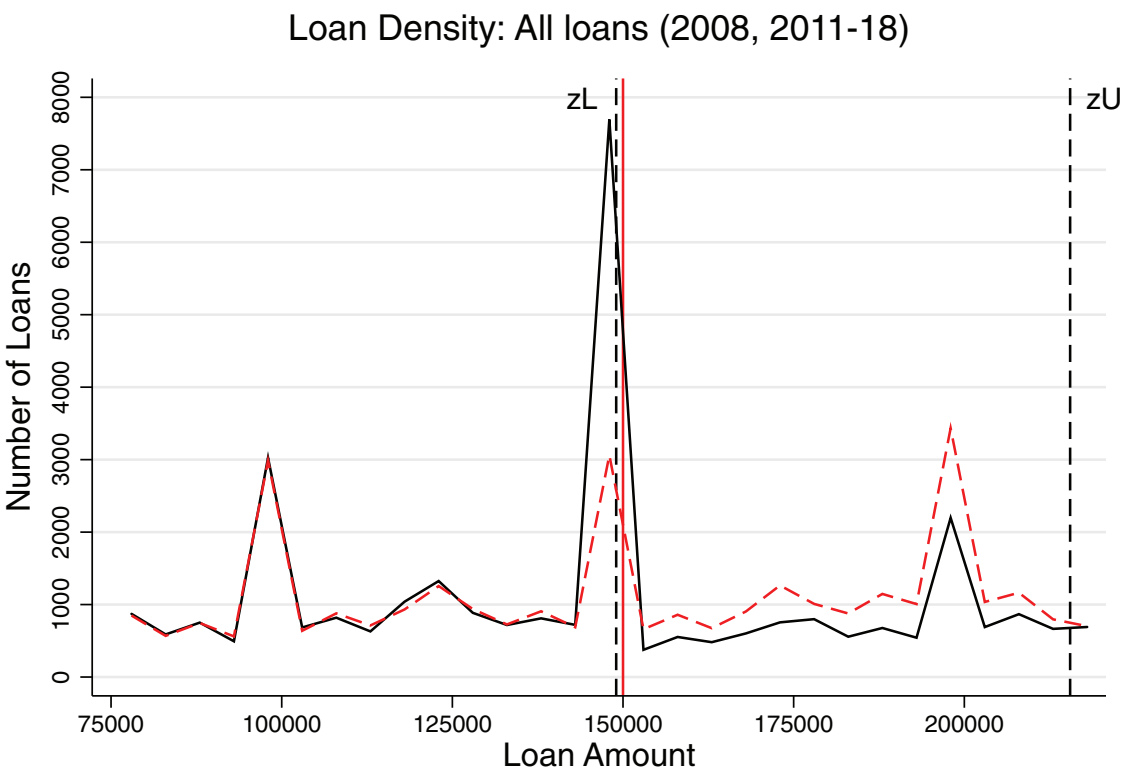



**Fig. 5.** Observed and counterfactual distributions.
*Notes:* This figure shows the observed and counterfactual density of loans. The solid line shows the observed number of loans, in $5,000 bins. The dashed line shows the counterfactual number of loans. The counterfactual is estimated for each notch separately by fitting a sixth-order polynomial with round number fixed effects to the empirical distribution using step size of 5000 and excluding data around the notch, as specified in Eq. (15). The dotted vertical lines mark the estimated excluded range $[d_L, d_U]$, where $d_U$ is the location of the marginal buncher. We calculate $\Delta D$ as $d_U - d_L$. Source: SBA.

The second panel shows estimates from years when the guarantee notch was binding. The estimates of the elasticity are between 4.5 to 5.2, depending on the polynomial used. Interpreted in dollar magnitudes, this means that a 3.8 percentage point change in the guarantee subsidy rate $(\Gamma)$ generates an approximate $70,500 in additional lending.

It is important to note that we estimate a reduced-form elasticity, which could be affected by optimization frictions. Optimization frictions are factors that prevent agents from locating at notch or kink points. For example, in labor supply estimates workers might be unable to alter their hours worked due to contractual arrangements, and in our context, projects might need a certain amount of capital. Notches are particularly useful in bunching estimators, relative to kinks, because in the absence of optimization frictions, theory predicts an excluded region to the right of the notch. Kleven and Waseem (2013) show methods to identify upper and lower bounds for the true structural elasticity. Under the admittedly strong assumption that the structural elasticity is homogenous, the reduced-form estimate is a structural elasticity. Otherwise, if there is heterogeneity in elasticities, the upper bound is represented by the response of the most sensitive individual.

We can place an additional restriction to identify the lower bound for the true structural elasticity, $\epsilon$. This approach requires identifying $\alpha$, or the share of individuals with sufficiently high adjustment costs that they are unresponsive to the notch. In this case, the term $\varepsilon_{D,\Gamma} = (1-\alpha)\epsilon$ is a lower bound for the true structural elasticity. We can use the share of individuals who do not optimize in a given year (i.e., the number of loans that are larger than $150,000 as a share of all loans in the dominated region) to estimate that $\alpha$. In years with a very high guarantee notch, we see approximately 40% of borrowers locating in the dominated region, suggesting that $\alpha \approx .4$. Thus we obtain a lower bound for the structural elasticity of approximately $\epsilon \approx 8.3$ (i.e., $5 = (1-0.4) \times \epsilon$). Intuitively, the reduced-form elasticity $\varepsilon_{D,\Gamma}$ is the observed elasticity attenuated by optimization frictions, $\alpha$. Therefore, the structural elasticity is greater than the reduced-form elasticity in the presence of frictions.

### 5.3. Bunching over time and placebo estimates

The observed amount of bunching varies over time with the size of the guarantee notch. Fig. 6 shows bunching at the guarantee notch for each year between 2008 and 2017. The figure groups years into three broad groups, years during which there is a high, low, or no guarantee notch. Between 2014 and 2017, the size of the notch was between 0.04 and 0.08 of the average expected guarantee benefit as a percentage of the loan principal. In 2008, and between 2011 and 2013, the notch was between 0.02

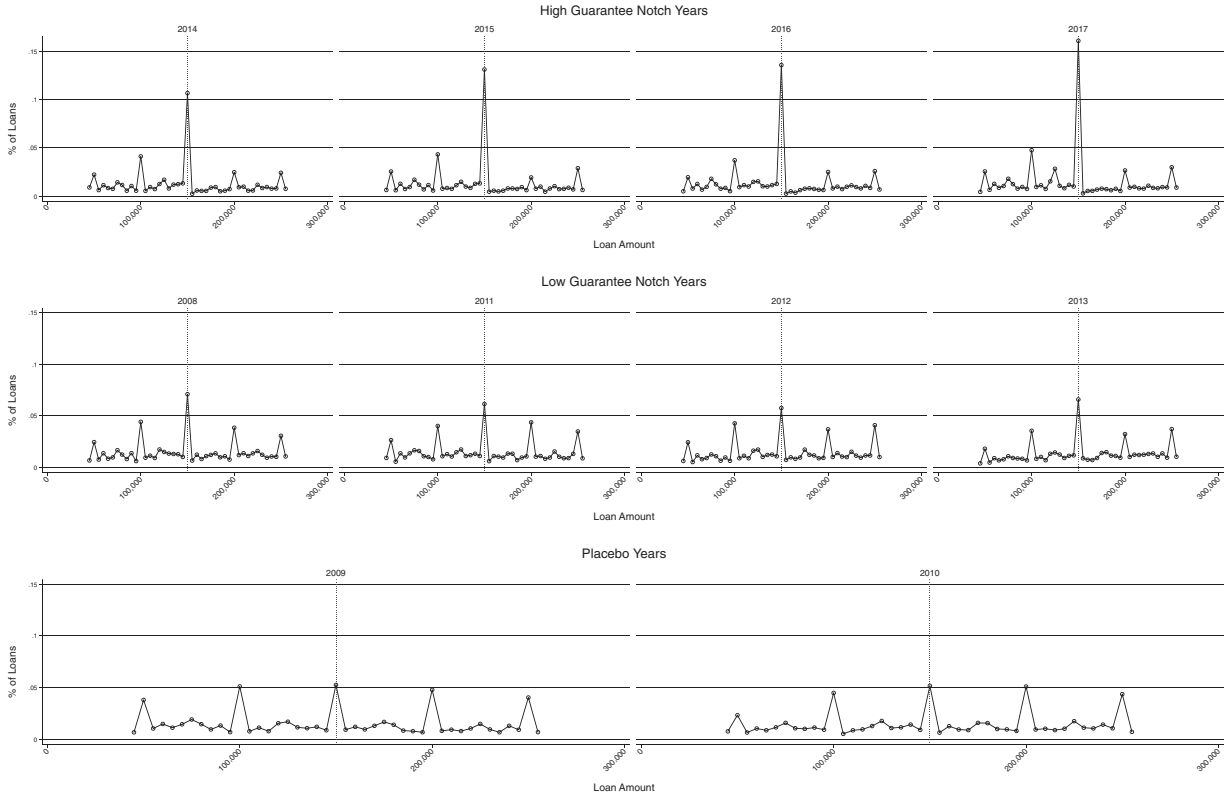



**Fig. 6.** Bunching at the guarantee notch by year.
*Notes:* This figure shows the fraction of loans made in discrete $2000 bins across the threshold by year. We divide the loans by years when the notch was either positive and above (high) or below (low) the median, or are nonexistent. Source: SBA.

and 0.03 of the average expected guarantee benefit as a percentage of the loan principal. In 2009 and 2010 the notch was eliminated as part of the ARRA.

We see a very close relation between the guarantee change and observed bunching at the threshold, defined as the difference in the share of loans between the observed and counterfactual density. In years with a large change in the guarantee, we see greater excess mass relative to years with a lower guarantee change at the notch. However, in years when the notch was eliminated (i.e., "placebo years"), there is no excess mass at the threshold.

Fig. 7 provides additional reduced-form evidence that this bunching is indeed driven by guarantees. The generosity of the guarantee across the notch has varied significantly over time, which allows us to explore dynamic aspects of the lending response. Consistent with the bunching being driven by loan guarantees, and not by any other factors changing across the threshold, we find higher excess mass in years when the difference in the guarantee across the threshold is greater. Fig. 7 shows the relation between share of excess mass at the threshold and the guarantee rate in each year. For this figure, we again use our reduced-form measure of excess mass: we observe some bunching at round number points, as is shown in Fig. A.6.

To account for this bunching, we calculate excess mass at the threshold relative to intervals of $50,000 between $50,000 and $300,000. The figure shows the amount of bunching occurring at the $150,000 threshold against the size of the guarantee change at the threshold between 2008 and 2017, in ten bins absorbing bank fixed effects. The left panel plots the share of excess mass and the change in the guarantee at the threshold. There is a striking linear relation between the share of excess mass and the guarantee rate. The right panel shows the relation between the share of excess mass and guarantee rates over time. The figure shows that the observed excess mass comoves with the guarantee rate, indicating a strong relation between the incentives to bunch and the amount of bunching.

Table 4 repeats the main analysis, showing estimates year by year. While estimates are relatively stable between 2008 and 2013, and similar in 2017, the estimates of $\varepsilon_{D,\Gamma}$ are about one-third the size of estimates in other years in 2014 and 2015. We see little excess mass in years when the notch was eliminated, and excess mass starts to grow sharply in 2014 when the guarantee notch becomes larger.

This growth in excess mass suggests the existence of adjustment frictions, where banks could take some time to increase credit supply. This can be seen in the right panel of Fig. 7. While there is a sharp jump in the guarantee notch between 2013 and 2014, approximately doubling from 0.033 to 0.077, the increase in excess mass is more gradual and increases year by year. The pattern translates to an initially lower elasticity, which increases to between 4.5 and 6 in 2017. Similarly, we observe some loans being

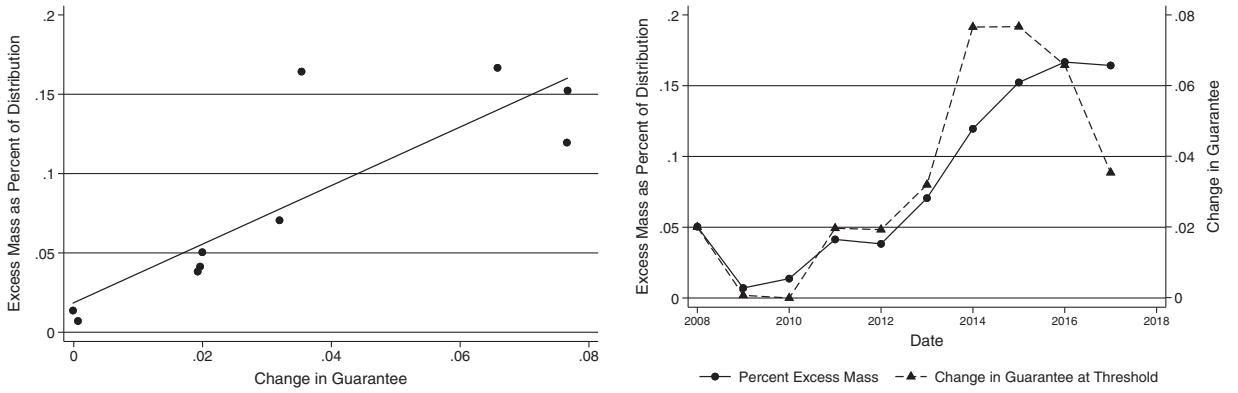



**Fig. 7.** Relation between size of notch and excess mass.
*Notes:* The figure on the left plots the share of excess mass against the size of the guarantee rate change at the $150,000 threshold. The excess mass at the $150,000 threshold is measured as the difference in the percentage of loans at the threshold relative to other round numbers. The share of excess mass is the estimated excess mass as a share of the total number of loans in the estimation range. The change in the guarantee rate is the change in the average expected guarantee benefit as a percentage of the loan principal. The figure on the right plots the share of excess mass and the size of the guarantee rate change at the threshold over time to show the tight correlation between the two measures. Both figures show that there is a positive correlation between the incentive to bunch (the size of the guarantee rate change) and the amount of bunching. Both graphs pool over all years 2008–2017 and control for bank fixed effects. Source: SBA.

**Table 4**
Excess mass and elasticity estimates, by Year.
This table shows elasticities for years in which a notch existed and estimates of the excess mass for the two years (2009 and 2010) in which there was no change in the guarantee rate at the 150,000 threshold. For this estimation, the stepsize = 500, the range was limited to 75,000-225,000, we included round number dummies for multiples of 1,5, 10, 25, and 50 thousand, and we used a polynomial of degree 6. The change in the guarantee rate ($\Delta\Gamma$) at the threshold for years in which a notch existed is computed as the weighted average of the average expected guarantee benefit as a percentage of the loan principal, where the weights correspond to the number of loans across years 2008, 2011–2017. Source: SBA.

| Year | Excess mass | $\Delta$D | $\Delta\Gamma$ | Elasticity |
|---|---|---|---|---|
| | *Placebo years - no notch* | | | |
| 2009 | 19.12 | 2,500 | 0 | NA |
| 2010 | 35.02 | 6,000 | 0 | NA |
| | *Years with notch* | | | |
| 2008 | 248.39 | 52,000 | 0.02 | 5.32 |
| 2011 | 151.81 | 40,500 | 0.02 | 3.36 |
| 2012 | 132.64 | 60,500 | 0.02 | 7.62 |
| 2013 | 199.91 | 71,500 | 0.03 | 6.41 |
| 2014 | 233.02 | 62,000 | 0.08 | 2.01 |
| 2015 | 457.83 | 55,500 | 0.08 | 1.61 |
| 2016 | 564.04 | 60,500 | 0.07 | 2.24 |
| 2017 | 1,386.12 | 69,500 | 0.04 | 5.47 |

made in the dominated region directly to the right of the threshold, suggesting that banks face optimization frictions when trying to adjust loan sizes. Therefore we estimate a reduced-form elasticity that is inclusive of adjustment costs rather than a structural elasticity.

The 2009 ARRA stimulus provides a placebo check. As part of the stimulus, the SBA temporarily raised the guarantee rate to 90% and waived fees in 2009. This effectively eliminated the guarantee notch at $150,000. It is immediately evident graphically that the lending response drops when guarantee notches are eliminated. The bottom rows of Fig. 6 shows the excess mass during years in which the notch was eliminated. Between 2009 and 2010, when guarantee rates were identical across the threshold, we do not observe any excess mass beyond round number bunching. The fact that excess bunching is only present in years when the guarantee rate is discontinuous assuages a potential concern that other factors might change discontinuously across the threshold.

### 5.4. Magnitudes

This section discusses the implied magnitudes of our estimates. The average guarantee subsidy rate over all years and for loans below $350,000 is 5.1% of loan principal. Thus a lender making a loan through the guarantee program receives a subsidy worth 5.1% of the loan size. The subsidy rate includes the expected reimbursement the lender will receive on any losses minus the guarantee fees ($\Gamma = \pi \cdot \gamma - \sigma$). Empirically, the guarantee subsidy generosity varies over years and loan size from -4%–when the guarantee fees outweigh the expected reimbursement–to 11.6%.

Our elasticity estimate suggests that an increase in one percentage point of the guarantee subsidy rate ($\Gamma$) for a given loan would generate an intensive margin response of $19,054 in additional lending. To increase the overall guarantee subsidy rate, the SBA could either increase the reimbursement portion ($\gamma$) or decrease the guarantee fees ($\sigma$). Increasing the reimbursement rate on a loan from 80% to 90% would increase the overall subsidy rate by $10\% \times \pi = 0.37\%$. The average charge-off rate over all years in our data is 3.7% and generates $8,002 in additional lending. This rate is based on the three-year cohort default rate. Decreasing the loan fee ($\sigma$) from 2.89% of loan principal (the average rate in 2008) to 0% (the rate in 2009) would increase the overall subsidy rate by 2.89% and generate $55,066 in intensive margin additional lending. Analyzed from the perspective of our model in Section 3.1, the elasticity suggests that additional lending has little impact on marginal default probabilities. Thus, lenders capture a relatively small portion of the subsidy.

**Table 5**
Effect of notch size on loan charge-off.
This table reports estimates of the notch size on charge-off probabilities. The column titles report the dependent variables. $(D > D^T)$ equals one if a loan size is greater than \$150,000. $\Gamma$ is the notch size. $(D = D^T)$ indicator equals one if the size of the loan is \$150,000. The sample is restricted to loan size between \$50,000 and \$225,000. Standard errors are reported in parentheses. The inclusion of fixed effects and controls is denoted beneath each specification. Source: SBA.

| | Probability of charge-off | | Percent charged-off | | Log of charged-off amount | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $(D > D^T)$ | −.008 | −.002 | −.005 | −.002 | −.085 | −.037 |
| | (.004) | (.012) | (.003) | (.010) | (.047) | (.142) |
| $(D > D^T) \times \Gamma$ | .034 | .100 | −.015 | .016 | .904 | 1.54 |
| | (.102) | (.083) | (.066) | (.063) | (1.15) | (.938) |
| $(D = D^T) \times \Gamma$ | .184 | .317 | .172 | .251 | 2.39 | 3.70 |
| | (.036) | (.092) | (.026) | (.070) | (.406) | (1.04) |
| Year FEs | X | | X | | X | |
| Size bin controls | | X | | X | | X |
| Bank FEs | | X | | X | | X |
| Year-month FEs | | X | | X | | X |
| Maturity FEs | | X | | X | | X |
| Number of observations | 40,751 | 40,751 | 40,751 | 40,751 | 40,751 | 40,751 |

These elasticities are on the higher end of estimates used for calibrations in Gale (1991) and consistent with elasticities used for model parameters in Lucas (2016). Lucas (2016) notes that supply elasticity is high in times of high levels of bank reserves and loose monetary policy. Overall, we argue that loan guarantees do indeed impact lending to small businesses by increasing loan volume.

### 5.5. Risk-shifting

A natural question is whether guarantees lead lenders to issue riskier loans. A higher portion of charged off dollars could induce lenders to be more lax in screening borrowers or to take fewer steps in monitoring borrowers and preventing defaults. One possibility is that the generosity of the guarantee rate pushes banks to lend to riskier borrowers (adverse selection) or deteriorates incentives to prevent charge-offs of loan applicants (moral hazard). Moral hazard and adverse selection on the part of the entrepreneur are unlikely in our context. Lenders, not borrowers, interface with the SBA programs. Borrowers rarely know that they are borrowing through the SBA program, and all changes in fees and reimbursement rates impact the bank directly, not the borrower. On the other hand, lenders might screen borrowers less thoroughly due to the guarantee.

We explore this question by exploiting temporal variation in the guarantee notch, testing whether banks shift loans more likely to be charged off to the notch when the guarantee benefit is higher. Table 5 shows estimates of variants of the following specification:

$$\pi_{i,t} = \alpha_i + \alpha_t + \alpha_m + \alpha_l + \delta \mathbf{1}(D > D^T) + \zeta \mathbf{1}(D > D^T) \times \Gamma + \xi \mathbf{1}(D = D^T) \times \Gamma + \nu_{i,t}, \quad (16)$$

where $\mathbf{1}(D = D^T)$ is an indicator of whether a loan is at the notch, $1(D > D^T)$ is an indicator of whether the loan is above the notch, and $\Gamma$ is the guarantee generosity. The outcome of interest is $\pi_{it}$, which is various measures of loan charge-off. Specifications include year or year-month fixed effects $\alpha_t$, lender fixed effects $\alpha_i$, maturity fixed effects $\alpha_m$, and loan size bin fixed effects $\alpha_l$. The main coefficient of interest is $\xi$, which captures the difference in charge-offs at the notch.

The results in Table 5 suggest that lenders indeed do shift riskier loans to the notch, where the guarantee rate is higher. The odd columns include only year fixed effects $\alpha_t$, while the even columns include year-month fixed effects as well as lender fixed effects $\alpha_i$, maturity fixed effects $\alpha_m$, and loan size bin fixed effects $\alpha_l$. In the first pair of estimates the dependent variable is an indicator of whether a loan is charged off, in the second pair of estimates the dependent variable is the percentage of the principal charged-off, while in the third pair of estimates the dependent variable is the log of the charged off amount.

Table 5 indicates that higher guarantee amounts are associated with higher charge-offs at the notch. All three dependent variables indicate higher levels of loan charge-off when the guarantee generosity is higher. A 10 percentage point increase in the generosity of the notch is associated with a 1.8 to 3.2 percentage point increase in charge-offs, a 1.7% to 2.5% increase in the amount of principal charged off, and a 0.23% to 0.37% increase in the amount charged off at the notch relative to the rest of the distribution. The estimates on the interaction term are statistically significant at the 0.01 level in all specifications. The table thus provides strong evidence that lenders are shifting risky loans to the notch when the guarantee generosity is higher.

Fig. 8 presents similar results graphically. Specifically, the top panel figure plots point estimates $\xi_t$ and a 95% confidence interval from the following specification:

$$\pi_i = \alpha_b + \alpha_y + \alpha_m + \alpha_l + \delta \mathbf{1}(D > D^T) + \zeta \mathbf{1}(D > D^T) \times \Gamma + \sum_{2008}^{2016} \xi_t \mathbf{1}(D = D^T) \times \Gamma + \nu_i. \quad (17)$$

Fig. 8 indicates that the patterns in the difference in charge-offs largely track the generosity of the notch in each year, which is shown in the bottom panel. The coefficients $\xi_{2014}$, $\xi_{2015}$, and $\xi_{2016}$ are particularly high when the size of the notch is greatest. We see very small and insignificant estimates of $\xi_{2009}$ and $\xi_{2010}$ when the notch was eliminated.

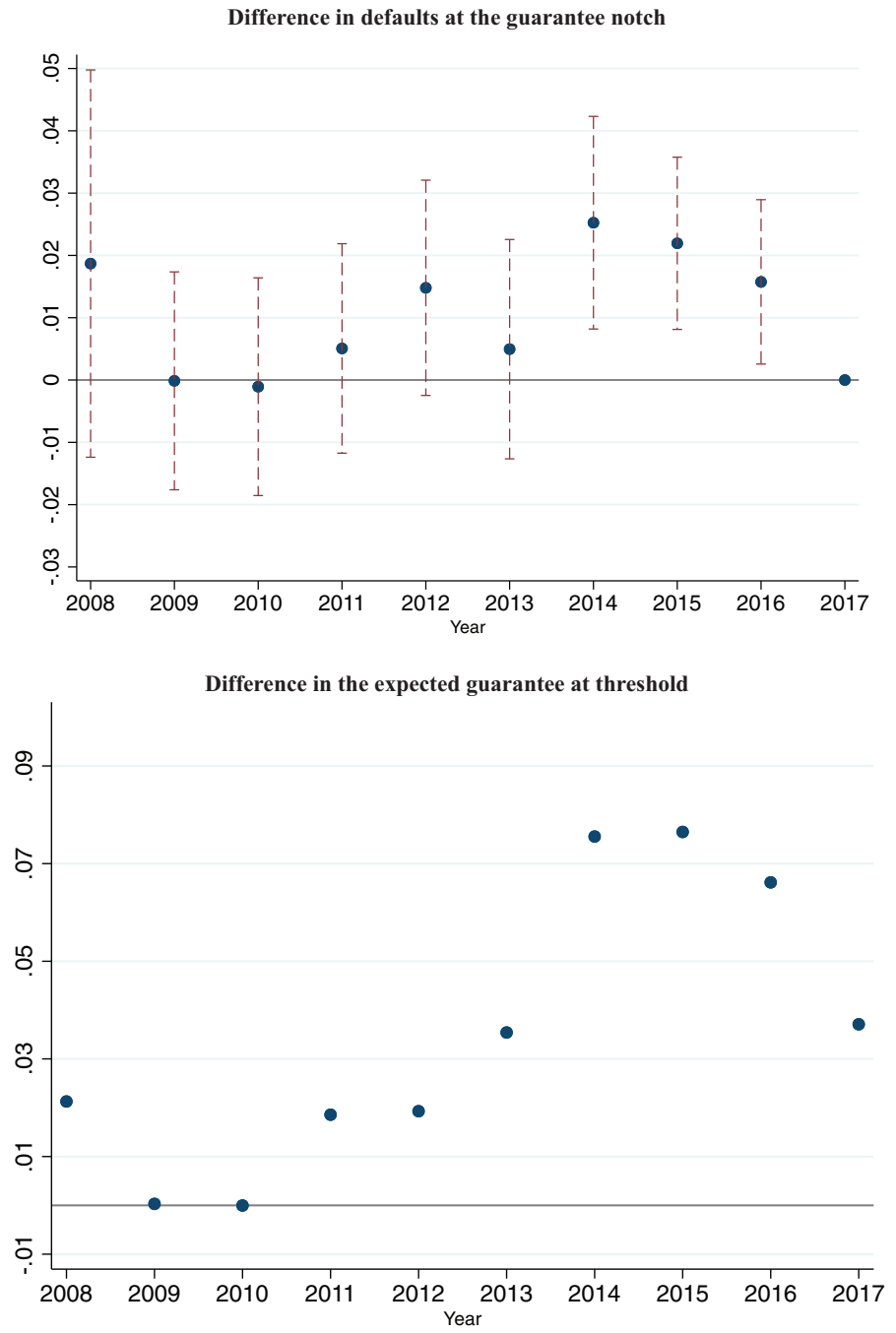



**Fig. 8.** Risk-shifting: charge-offs at notch.
*Notes:* The top panel plots the coefficient $\xi_t$ on the interaction terms between whether a loan is at the guarantee notch and year indicators in Eq. (17), along with a 95% confidence interval. The dependent variable is an indicator of whether a loan is charged off. The baseline is 2017. The bottom panel shows the magnitude of the expected benefit at the guarantee notch. Source: SBA.

## 6. Alternative channels, robustness, and placebo estimates

### 6.1. Demand and supply elasticities

One concern is that our estimates do not identify lenders' elasticity of supply to the guarantee rate but rather identify borrowers' elasticity of demand. It is in theory possible that guarantees are passed through to borrowers in the form of lower interest rates. Specifically, borrowers might be more likely to apply for a smaller $150,000 loan if the guarantee is passed through via a lower interest rate or lower risk standards. However, there are several institutional details that make a demand

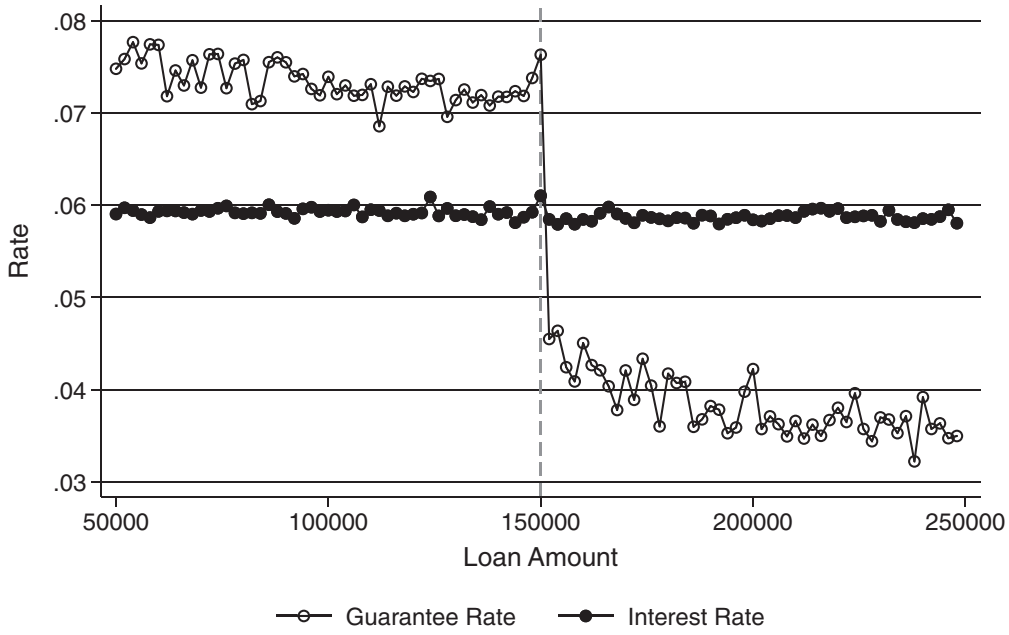



**Fig. 9.** Average interest rate and guarantee rate across the threshold.
*Notes:* This figure shows interest rates and guarantee rates in discrete $2000 bins across the threshold. While the guarantee rate drops dramatically at the threshold, the interest rate remains flat. The guarantee rate is the average expected guarantee benefit as a percentage of the loan principal. The graph pools over all years 2008–2017. Source: SBA.

channel unlikely. As noted earlier, lenders are unable to issue multiple loans to the same borrower under the SBA program, making manipulation of the notch unlikely. Furthermore, borrowers must have exhausted all other financing options to qualify for an SBA loan, which rules out the possibility that banks or borrowers are topping up their SBA loans with additional private funding. Indeed, the eligibility criteria listed on the SBA website specifically states that to qualify for a 7(a) loan, "the business cannot get funds from any other financial lender." The observed data are also inconsistent with this demand side hypothesis. We find that a negligible portion (0.03%) of loans are categorized as revolving debt, i.e., a line of credit that can be drawn down by the borrower, and could also lead to demand-driven manipulation of the notch.

Despite the fact that institutional details make this demand channel unlikely, we verify whether the notch induces borrowers to bunch at the threshold by observing whether interest rates or ex-post charge-off rates (a measure of borrower risk) change discretely at the threshold. Fig. 9 shows average interest rates and the guarantee notch. Interest rates evolve smoothly despite the sharp guarantee notch. Fig. A.7 provides some insight as to why this could be the case–the majority of loans are priced at the cap on each side of the threshold.

Fig. A.8 shows that other factors trend smoothly across the threshold. Interest rates, revolving loan status, charge-offs, and loans terms all evolve smoothly, which suggests that the generosity in the guarantee is not passed on to the borrower through either an intensive margin interest rate effect or an extensive margin rationing effect. The figure implies that borrowers have no incentives to bunch at the threshold because requesting smaller loans to bunch at the notch only gives them less capital with no added benefits. Given this lack of incentives to bunch from the perspective of the borrowers, it is unlikely that the bunching is demand driven.

It is also possible in theory that borrowers request smaller loans than they otherwise would have if they believed that bunching at the notch improves their odds of getting the loan approved. If this is the case, this is still interpretable as a supply elasticity since it is operating through a supply side mechanism: the approval rate. If the supply side was not reducing credit supply to the right of the notch, borrowers would not modify their loan requests.

### 6.2. Competition and loan substitution

#### 6.2.1. Loan substitution

One potential concern is that we are not measuring a supply elasticity but rather a substitution elasticity (i.e., the loan guarantee is not increasing total credit supply, but rather incentivizing banks to shift loans from their SBA small business portfolio into the non-SBA portfolio or vice versa). Such within-bank substitution would generate a discontinuity in the number of loans originated at the $150,000 size cutoff. While this channel would not generate excess mass at the $150,000 notch, it could generate spurious missing mass to the right of the notch if banks place low-guarantee loans in their non-SBA portfolio.

To assuage a concern that spurious missing mass can confound our elasticity estimates, we estimate and compare elasticities on loans originated by a subsample of lenders that do and do not specialize in making SBA loans.

**Table 6**
Estimates split by number of banks and SBA share.
This table reports estimates of excess mass and the main elasticity estimates by subsamples of loans that are and are not originated by lenders that specialize in SBA lending (top two panels) and by subsamples of loans originated in counties with more or less operating lenders (bottom two panels). The share of SBA lending in the top two panels is computed from calculating the share of SBA loans relative to a lender's overall small business lending. The estimation restricts the sample to loans to be of size between $75,000 to $225,000, uses the step size of 500, and includes round number dummies for multiples of 1,5, 10, 25, and 50 thousand. The degree of the polynomial used in the estimation is denoted in the second column. The change in the guarantee rate ($\Delta\Gamma$) at the threshold for years in which a notch existed is computed as the weighted average of the average expected guarantee benefit as a percentage of the loan principal, where the weights correspond to the number of loans across years 2008, 2011–2017. Standard errors are shown in parentheses. Source: SBA and FDIC SDI.

| Year | Polynomial | Excess mass | $\Delta D$ | $\Delta\Gamma$ | Elasticity | Excess mass | $\Delta D$ | $\Delta\Gamma$ | Elasticity |
|---|---|---|---|---|---|---|---|---|---|
| | | SBA share $>$ 60% | | | | SBA share $\leq$ 60% | | | |
| 2008, 2011–2017 | 6 | 2,336 | 69,500 | 0.038 | 5.087 | 2,413 | 55,500 | 0.038 | 3.244 |
| | | (20.63) | (3,286) | – | (0.466) | (40.18) | (7,254) | – | (0.942) |
| | No. obs. | | 8931 | | | | 24,333 | | |
| | | SBA share $>$ 80% | | | | SBA share $\leq$ 80% | | | |
| 2008, 2011–2017 | 6 | 2,231 | 69,500 | 0.038 | 5.087 | 2,518 | 55,500 | 0.038 | 3.244 |
| | | (20.41) | (3,299) | – | (0.465) | (41.04) | (7,489) | – | (0.975) |
| | No. obs. | | 7,958 | | | | 25,306 | | |
| | | Unique banks $>$ 3 | | | | Unique banks $\leq$ 3 | | | |
| 2008, 2011–2017 | 6 | 4,363 | 66,000 | 0.038 | 4.589 | 383 | 53,500 | 0.038 | 3.015 |
| | | (46.23) | (6,149) | – | (0.852) | (9.18) | (6,889) | – | (0.887) |
| | No. obs. | | 28,851 | | | | 4,413 | | |
| | | Unique banks $>$ 7 | | | | Unique banks $\leq$ 7 | | | |
| 2008, 2011–2017 | 6 | 3,931 | 70,000 | 0.038 | 5.161 | 818 | 60,500 | 0.038 | 3.855 |
| | | (37.86) | (5,819) | – | (0.812) | (17.00) | (7,858) | – | (1.031) |
| | No. obs. | | 24,509 | | | | 8,755 | | |

A number of lenders, such as Live Oak Bank, specialize in making SBA-guaranteed loans and offer few, if any, other products. Thus, if the elasticity estimates between specialized and nonspecialized lenders are similar, it implies that it is unlikely that lenders shift loans between SBA and non-SBA products.

To identify lenders who specialize in SBA lending, we link SBA lenders to Call Report data and compute the total share of SBA loans originated by each lender. Next, we merge the SBA data set with quarterly Statistics on Depository Institutions (SDI) data from the FDIC to capture non-SBA loans. We match the majority of banks in our data (including federal credit unions) at an overall rate of 83% and a rate of 96% conditional that Call Report data exist (prior to Q1 2010, SDI reports were only provided yearly in Q2). The SDI data record the total number and amount of small business loans outstanding at a quarterly level per institution and further split small business lending into categories of loan size and purpose. We specifically look at small business commercial and industrial loans under $1 million since these are most comparable to those provided through the 7(a) program. We also aggregate the SDI statistics to the yearly level. Appendix A.3 provides further information on the FDIC SDI data and a description of how we compute the SBA loan share by lender.

The top two panels of Table 6 show sample splits by lenders that do and do not specialize in SBA lending. Plots of the estimated counterfactual density for both splits are in Fig. A.13. The first panel splits lenders by whether the share of SBA loans is above or below 60% of their entire loan portfolio, while the second panel splits lender by above and below 80% share. The elasticity estimates are slightly higher at SBA specialized lenders, but overall the estimates are very similar. Thus, we do not find evidence that our results are biased by lenders substituting loans between SBA and non-SBA products.

In addition to comparing elasticities, we directly test whether banks that specialize in SBA lending are more likely to substitute non-SBA for SBA loans when the guarantee generosity is higher. If higher guarantees incentivize banks to shift their small business portfolio to SBA loans, we would expect this effect to be concentrated among banks with higher propensity to issue SBA loans relative to other small business loans. We explore this in the appendix by comparing differential responses between high- and low-SBA share lenders when guarantee rates were increased during 2009 and 2010. Table A.4 shows estimates of the following specification:

$$D_{i,t} = \alpha_i + \alpha_t + \delta \mathbf{1}(\text{Treat}) + \zeta \mathbf{1}(\text{Treat}) \times \theta_i + \varepsilon_{i,t}, \qquad (18)$$

where the outcomes include the log of total, non-SBA, or SBA loan amounts; $\mathbf{1}(\text{Treat})$ is an indicator that equals one for years 2009 and 2010; $\theta_i$ is a bank-specific share of the amount of SBA lending relative to its overall small business lending portfolio in 2008; and $\alpha_i$ and $\alpha_t$ are bank and year fixed effects. Since the reimbursement rate increased to 90% on both sides of the $150, 000 threshold in 2009 and 2010 due to the ARRA stimulus, the estimated coefficients $\delta$ and $\zeta$, respectively capture the effect of increased guarantees on the composition of small business lending portfolio and the differential response for banks with higher pre-ARRA propensity to issue SBA loans.

Table A.4 shows that while banks with higher pre-ARRA share of SBA lending increased SBA lending more in response to higher guarantees, the effects on non-SBA and total loan supply are not statistically significant. Fig. A.9 illustrates this finding graphically by plotting $\zeta_t$ from the following equation for total and non-SBA and SBA small business loans:

$$D_{i,t} = \alpha_i + \alpha_t + \delta \mathbf{1}(\text{Treat}) + \sum_{2008}^{2016} \zeta_t \mathbf{1}(\text{Year} = t) \times \theta_i + \varepsilon_{i,t}. \tag{19}$$

Consistent with the results in Table A.4, while banks with higher propensity to issue SBA loans differentially increase SBA loan supply in 2009 and 2010, with $\zeta_{2009}$ and $\zeta_{2010}$ being positive and significant for SBA loans, the effect on non-SBA loans are not statistically different from zero. These results confirm that within-bank substitution between SBA and non-SBA loans are unlikely.

#### 6.2.2. Competition

A notch can incentivize borrowers to smooth the lenders' bunching behavior through borrowing from other sources. To the extent that a notch leads borrowers to seek funds from other sources, this can mitigate the credit supply effect. The institutional detail suggests that this is unlikely. SBA 7(a) loans typically carry higher interest rates than most other loan products, making it unlikely that borrowers would seek SBA loans if other financing options are available. Moreover, the SBA requires that lenders document and verify that a borrower passed the credit elsewhere requirement, which demonstrates that a borrower has "exhausted" all options for getting funds and cannot obtain funds without undue hardship.

While the SBA loans are intended to serve borrowers that cannot obtain loans elsewhere, it is still possible that this test is ineffective or poorly enforced. To explore this channel, we conduct sample splits by the number of banks operating in a borrower's county. In geographic areas with fewer operating banks, it can be more difficult for firms to access other forms of credit because the market is concentrated. Thus, if estimates are similar across areas with varying bank competition, we can infer that credit availability in a local market plays a small role in how lenders respond to changes in the guarantee rate.

The bottom two panels of Table 6 report the results. The first panel splits the sample by loans originated in areas where the number of banks is above or below three, while the second panel splits the sample by areas where the number of banks is above or below seven. While the estimates in counties with fewer banks are slightly lower relative to counties with more bank competition, we still observe significant excess mass and large elasticities between three and five in counties with fewer banks. This suggests that we see similar bunching effects in less competitive markets.

The top two panels of Table 6 report that we see similar elasticities for specialized lenders that are very likely to be compliant with the credit elsewhere test. Lenders can be excluded from the guarantee program if they repeatedly fail to verify credit elsewhere tests. Since exclusion from the program is extremely costly for lenders that specialize in making SBA loans, they are very likely to be compliant with the credit elsewhere test. Thus, this result supports the idea that there is a significant lending supply response from lenders who are compliant with the credit elsewhere test.

### 6.3. Alternative ranges

Table A.5 varies the range used in the estimation. We vary the loan sample range and bin size. The first column denotes alternative loan size ranges, while the top row denotes alternative bin sizes. The elasticities remain large and significant–between three and seven–when using alternative ranges and bin sizes, similar to those reported in Table 4 when we vary the bin size. There is some variation in the elasticity estimates stemming from varying the estimation range. There are two factors that make the estimates somewhat sensitive to the choice of estimation range.

First, there are two additional thresholds to the left and the right of the guarantee notch, which constrain our ability to extend the estimation region further and cause estimates to change in the vicinity of these notches. First, there is an interest rate cap notch at \$50,000. For loans below \$50,000, lenders can charge a one percentage point higher interest rate. Second, there is a collateral notch at \$350,000. Lenders are required to collateralize the loan to the maximum extent possible up to the loan amount when they borrow over \$350,000. If business assets do not fully secure the loan, the lender must take equity in the principal and/or personal real estate as collateral. This leads to significant bunching at \$350,000. We thus avoid ranges near the other policy notches.

Second, much of the variation in the calculated elasticities stemming from varying the upper bound of the range is driven by the estimation procedure itself. We calculate the elasticity with the upper bound when the terminal condition of the estimation is reached. In other words, the mass of $\hat{B}$ and $\hat{M}$ are in a given tolerance range. Given the iteration in the estimation procedure, this sometimes leads to the result that $\hat{M} > \hat{B}$. This is because we do not assign the upper bound as the value that best equalizes $\hat{M}$ and $\hat{B}$ but assign the $d_U$ at which the estimation is terminated because the tolerance threshold is reached. This terminal $d_U$ can in some cases be greater than the $d_U$ that equalizes $\hat{M}$ and $\hat{B}$. For most of the specifications–including our main preferred spec–this does not happen (i.e., the terminal $\hat{M} = \hat{B}$ is within a 0%-5% margin). However in the rare cases when it does happen, this can cause the elasticities to jump.

We further explore alternative estimation ranges graphically. Fig. 10 shows point estimates of the elasticity using various ranges, along with a 95% confidence interval. The estimates are generally centered around 5, with a range between 2.5 and 11. The estimates generally increase when we expand the range. Appendix Fig. A.10 shows that the estimates are more sensitive to varying the upper limit rather than the lower limit.

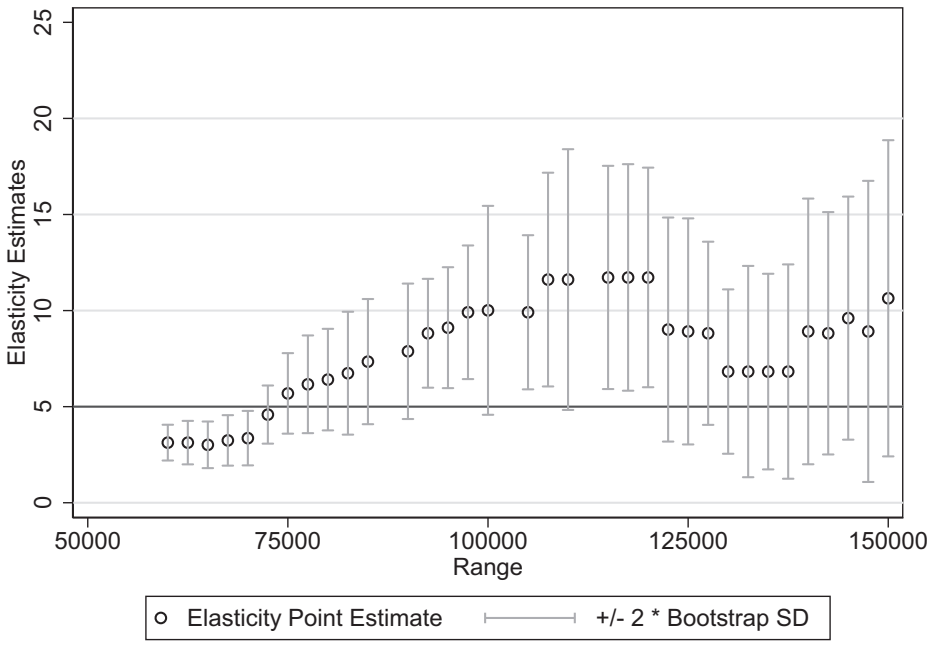



**Fig. 10.** Estimates varying estimation range.
*Notes:* This figure shows elasticity estimates varying the starting and ending range around the $150,000 threshold. A sixth-order polynomial is used. The bars show a 95% confidence interval. Standard errors are computed using a bootstrap. Source: SBA.

### 6.4. Additional robustness

We show in the appendix that the elasticity estimates are robust to a number of alternative specifications. Our main estimates use a polynomial of degree six to estimate the counterfactual loan distribution, the sample of loan size between $75,000 and $225,000, and a step size of $500 when iterating through the estimation routine to find the upper limit of the excluded zone. Tables A.6 tests the sensitivity of our elasticity estimates by varying key parameters. In Table A.6, we vary the polynomial (top panel) to degree five and seven while keeping the step size constant and vary the step size while keeping the polynomial constant. The elasticity estimates are robust to the choice of polynomial and do not exhibit a specific direction of bias (smaller or larger) when we increase the polynomial degree.

We explore heterogeneity in elasticity estimates by borrower's project location and sector characteristics. Appendix Table A.7 shows that the elasticity is larger in project areas with high local bank competition relative to areas with low bank competition, where bank competition is measured using the Herfendahl-Hirschman Index (HHI). This result is consistent with the results found in Table 6 that shows heterogeneity by SBA market concentration. Heterogeneity by sector characteristics show that the elasticity is larger for the borrowers in high exit, services-producing, and tradable sectors relative to those in low exit, goods-producing, and nontradable sectors. One interpretation of this result is that lenders may want to bunch more when lending to borrowers that operate in risky sectors with higher chances of default or in sectors that traditionally require high initial capital. Appendix Table A.8 reports elasticity estimates by industry for the top five industries in terms of the share of SBA loans provided to a given industry relative to the total SBA loans.

Our assumption of risk neutrality does not qualitatively affect the bunching estimation procedure, but it does affect the underlying structural elasticities estimated. Under risk neutrality, a change in the reimbursement rate $\gamma$ is equivalent to a change in the fee $\sigma$. However, if the banks are risk averse, a decrease in reimbursement rate $\gamma$ does not linearly correspond to an increase of participation fee $\sigma$, and banks' expected subsidy rate can no longer be represented as a netted quantity $\Gamma(D_{ij})$, which drops discretely at threshold $D^T$. Even if the optimal $D_{ij}$ remains unchanged, the slope of the indifference curve could potentially change–it will be steeper as banks become more risk averse. This will reduce the size of the marginal bunching loan relative to the modeled scenario with risk-neutral banks. As a consequence, we would expect more bunching for more risk averse banks. Intuitively, staying above the threshold and getting a lower guarantee rate is a greater potential loss to more risk-averse banks.

## 7. Concluding remarks

The efficiency of federal credit guarantees depends crucially on how responsive the lending supply is to subsidies. Specifically, the marginal change in costs per dollar of lending generated affects the elasticity of loan supply to the guarantee. This paper uses notches in SBA lending rules to provide the first empirical estimates of credit supply response to guarantees. We find that supply is responsive to loan guarantees-significantly more loans are disbursed below a threshold where guarantees are higher

and we find that this bunching is stronger in years when the discontinuity in guarantee rates is greater and disappears when the discontinuity is temporarily eliminated. We translate these estimates into an elasticity of credit supply with respect to a loan guarantee of approximately five, which is on the higher end of those used in previous studies. We thus conclude that lending is highly sensitive to loan guarantees, and these programs have significant potential to increase lending levels when borrowing is inefficiently low.

While we have shown that lending supply is responsive to guarantee rates–a key parameter when considering the welfare effects of federal credit programs–important questions remain unanswered. Perhaps most importantly, the efficiency of loan guarantees ultimately rests on the efficiency of the rate of return on investments made by marginal loans and whether this is greater than the risk-free rate. Moreover, federal credit programs can have allocative effects, transferring credit from one rationed group to another. Future work should attempt to study both the allocative effects of federal credit programs and the return of loans being made under these programs.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jfineco.2020.08.008.

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